

FAA AACE Grant for FAA Development of Reliable Modeling Methodologies for Fan Blade Out Containment Analysis

ASU Research Team

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Technical Review Meeting

May 6th, 2010

Arizona State University

Tempe, AZ 85287

Overview and Objectives of Phase 3: Experimental Program

Phase 2

- Uniaxial tension tests
- Picture Frame Tests
- Friction Tests
- Static Ring Tests

Current Program

- Single Yarn Pull-out Tests
- Kevlar 49 Fabric High Speed Tests
- Kevlar 49 Single Yarn Tests
(Quasi-Static and High Speed)
- Shear Tests
- Biaxial Tests

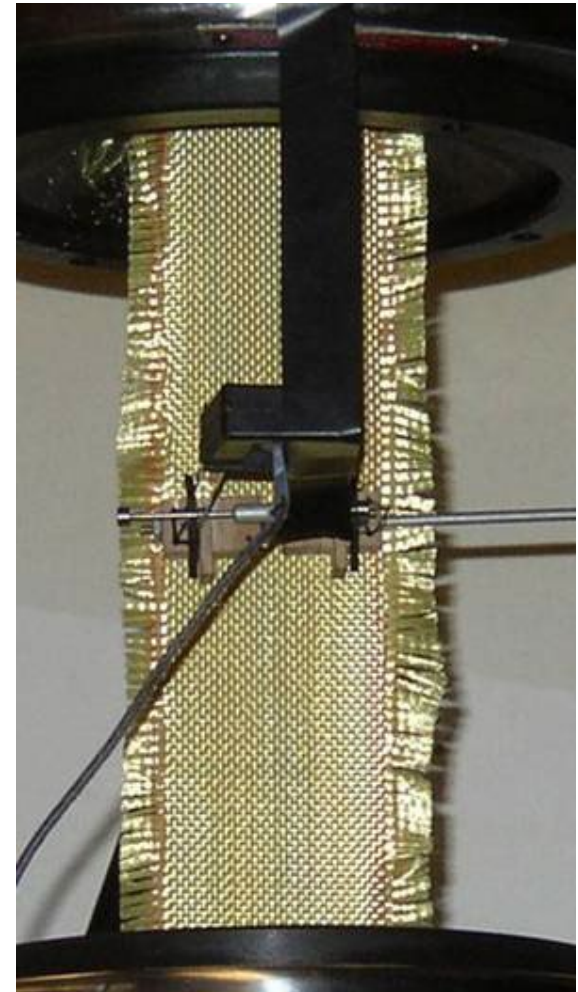
Outline

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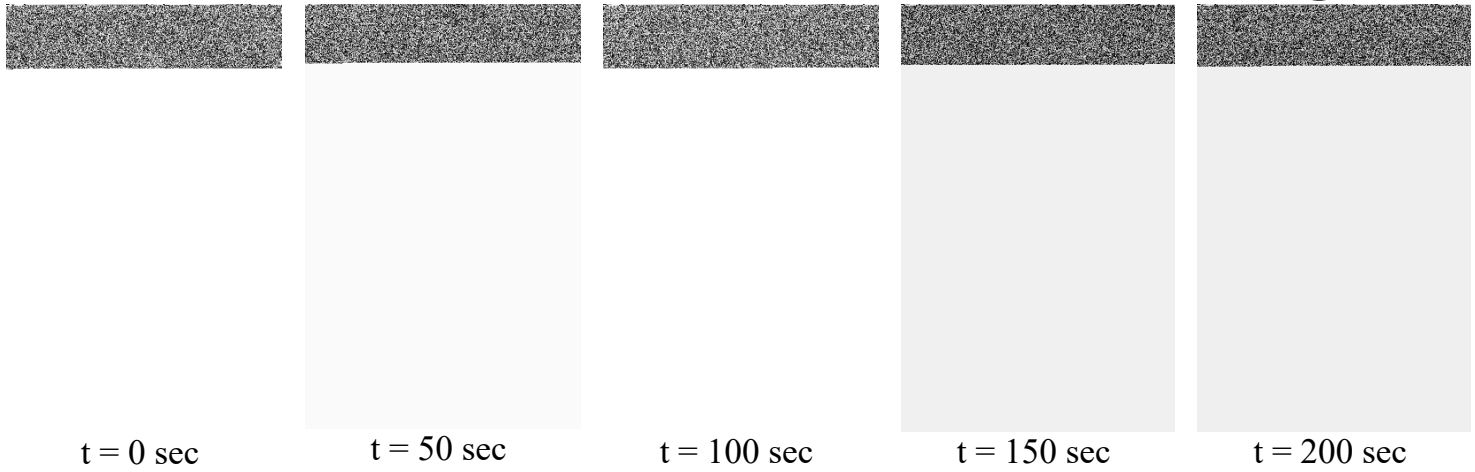
- **Uniaxial Tensile Test**
 - Image Analysis: Full-field displacement
 - Stress-strain curves for models (v1.3)
- **Biaxial Test**
 - Effective Poisson's Ratio
 - Pre-load effect
- **Modal Analysis and High Speed Test**
- **Picture Frame Test**
 - Shear behavior
 - Size effect
- **Pull-out Test: friction and preload effect**
- **Single Yarn (Virgin Yarn) Quasi-static Test**

Uniaxial Tension Tests

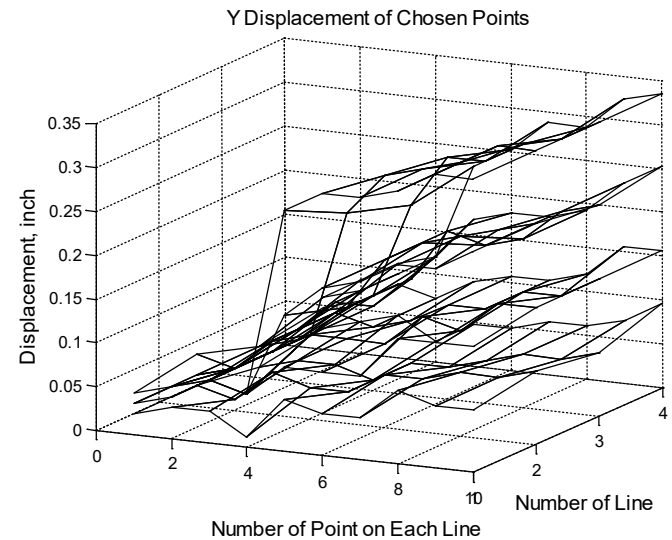
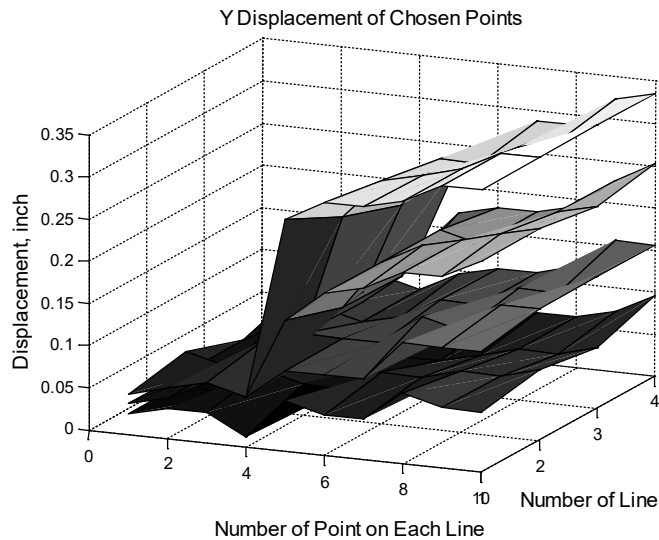
- Uniaxial tension tests of the textile fabrics performed in accordance with pertinent ASTM Standards.
- Properties that are obtained from this test include:
 - Tensile modulus of elasticity
 - Poisson's ratio in tension
 - Ultimate tensile strength, tensile strain.
 - Total Energy absorbed
 - Slack



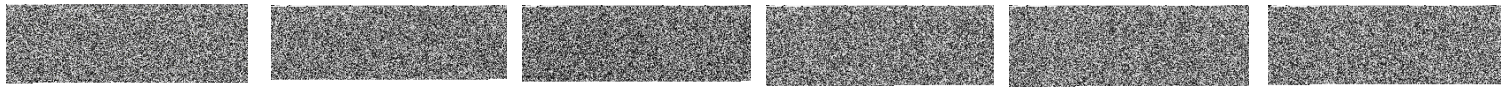
Quasi-static Uniaxial Tensile Testing



Images captured during loading process



Full-field Y displacement of Kevlar specimen at time of 50, 100, 150 and 200 s



$t = 0 \text{ sec}$

$t = 30 \text{ sec}$

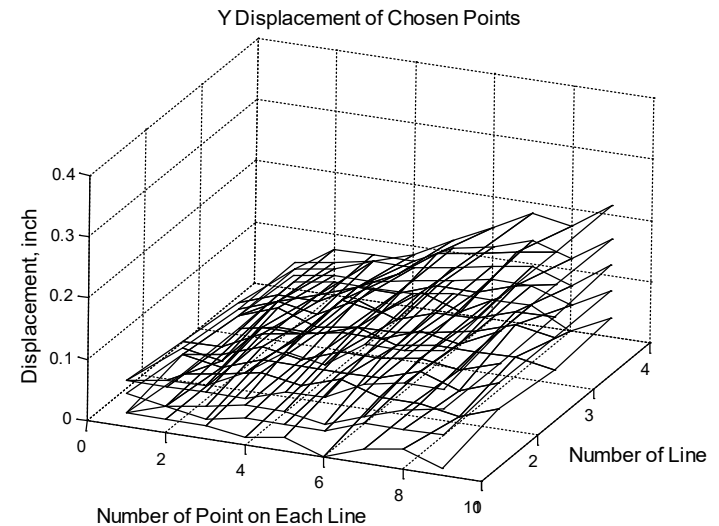
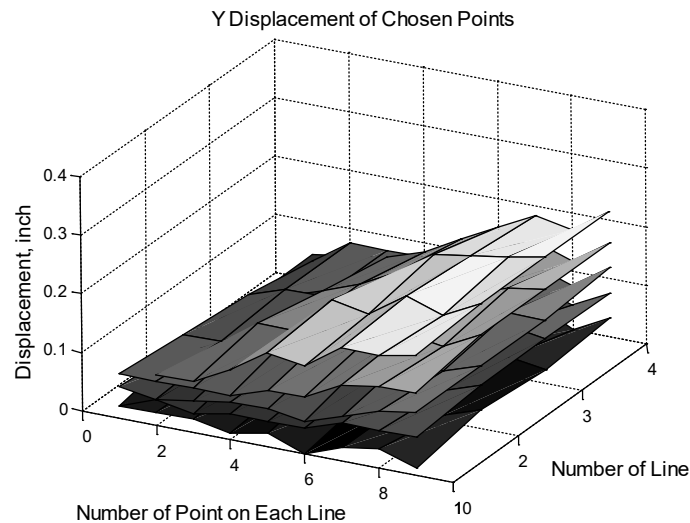
$t = 60 \text{ sec}$

$t = 90 \text{ sec}$

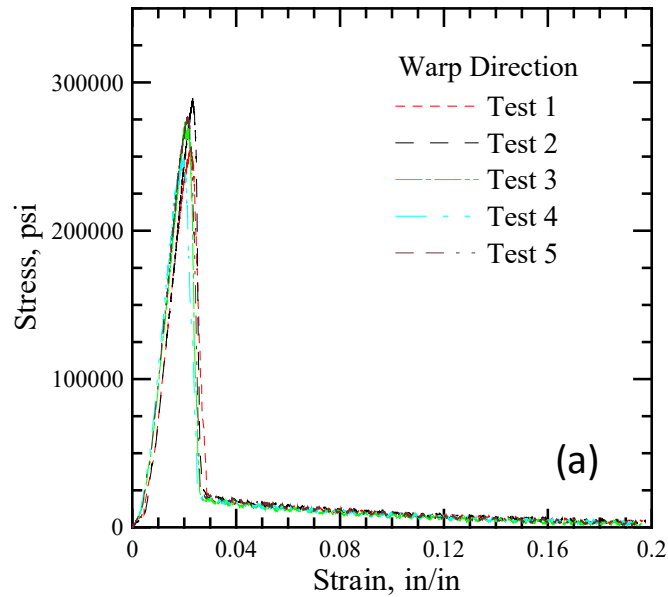
$t = 120 \text{ sec}$

$t = 150 \text{ sec}$

Images captured during loading process

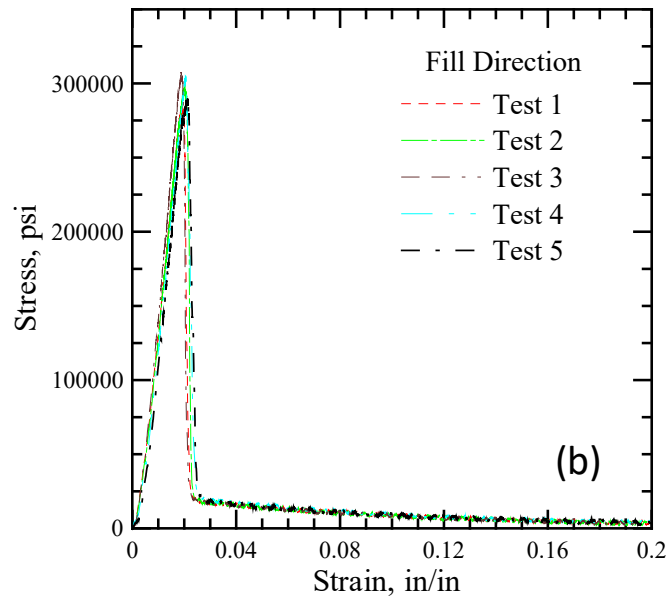


Full-field Y displacement of Kevlar specimen at time of 30, 60, 90, 120 and 150 s



Summary of 8"x1" specimens in warp direction

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	257247	16608094	4827	0.0225
2	289379	17021658	4899	0.0233
3	273561	17191336	4627	0.0204
4	250057	17065180	4512	0.0195
5	277710	17628840	4991	0.0216
Average	269591	17103021	4771	0.0215
Std. Dev.	15868	366500	197	0.0015



Summary of 8"x1" specimens in fill direction

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	284806	16896553	4589	0.0193
2	299015	16686524	4825	0.0202
3	307532	17513212	5031	0.0206
4	308640	18138606	4717	0.0189
5	290157	17575759	4900	0.0211
Average	298030	17362131	4812	0.0200
Std. Dev.	10496	579781	169	0.0009

Stress-strain curves of 8"x1" specimens in (a) warp and (b) fill directions

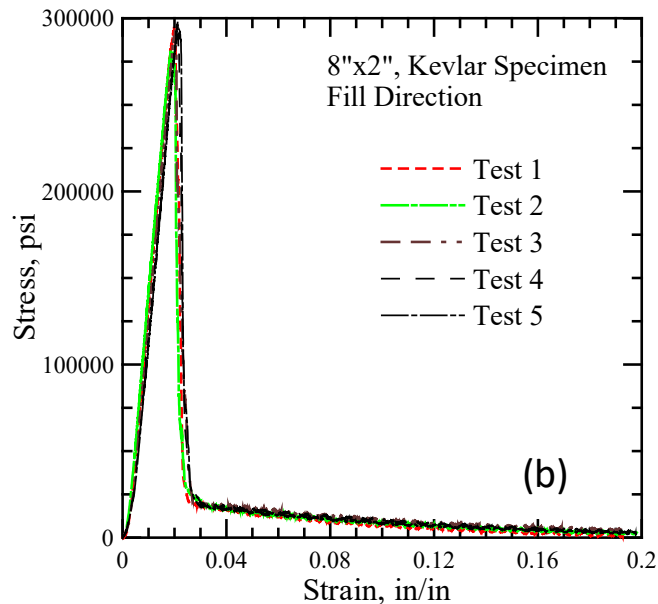
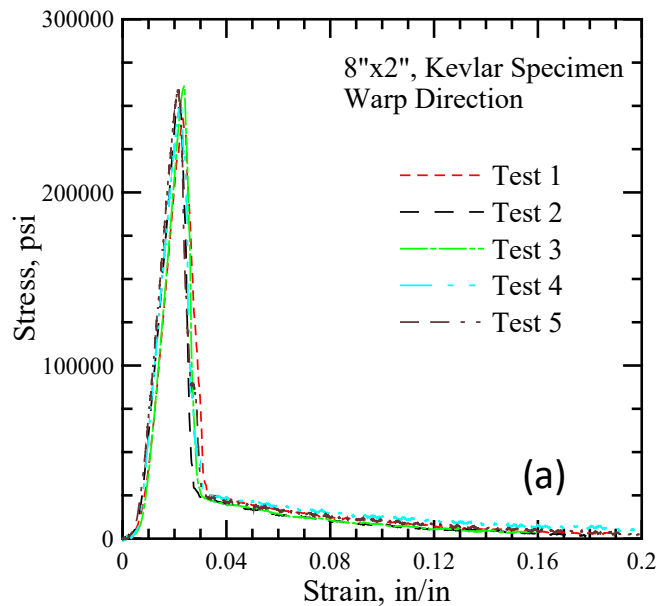
Summary of 8"x2" specimens in warp direction

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	242883	16239106	4798	0.0231
2	250878	17120831	4253	0.0208
3	262851	17286792	4232	0.0238
4	250424	16846784	5231	0.0224
5	260286	17462721	4951	0.0213
Average	253464	16991247	4693	0.0223
Std. Dev.	8102	477758	440	0.0012

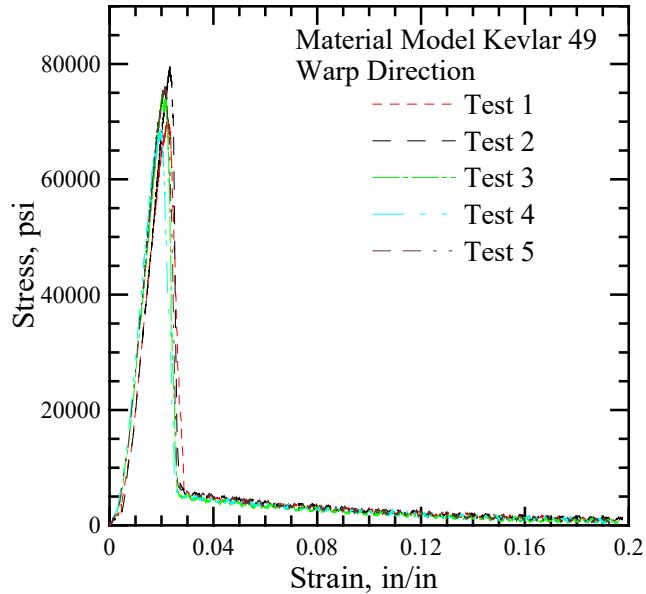
Summary of 8"x2" specimens in fill direction

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	294611	17555369	4622	0.0198
2	281243	17193379	4846	0.0186
3	293236	17111610	5097	0.0201
4	293003	16522925	5072	0.0210
5	298017	16559192	4927	0.0212
Average	292022	16988495	4913	0.0201
Std. Dev.	6349	441457	192	0.0010

Stress-strain curves of 8"x2" specimens in (a) warp and (b) fill directions

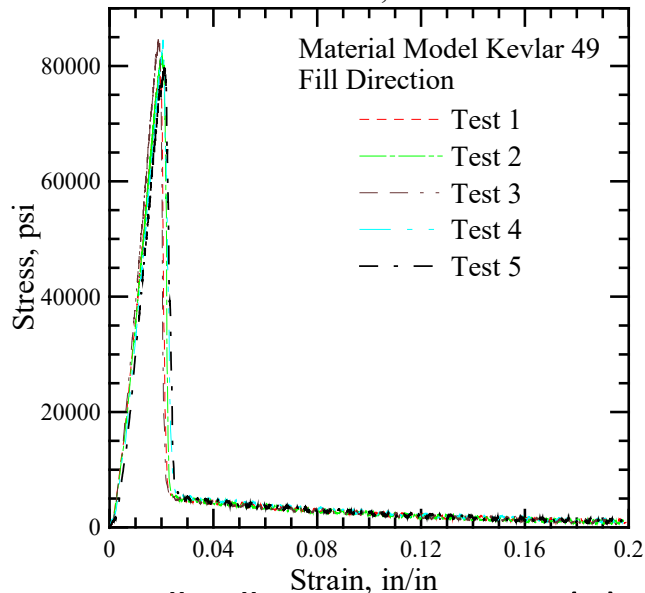


Stress-Strain Curves used in Material Model



Summary of 8"x1" specimens in warp direction in the material model

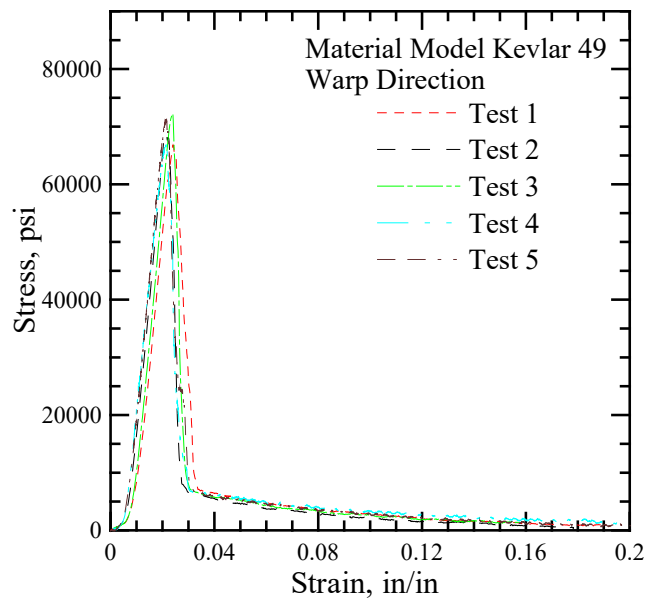
Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	70720	4571045	1327	0.0225
2	79553	4678867	1347	0.0233
3	75204	4790262	1272	0.0204
4	68743	4606335	1240	0.0195
5	76345	4895609	1372	0.0216
Average	74113	4708424	1312	0.0215
Std. Dev.	4362	134075	54	0.0015



Summary of 8"x1" specimens in fill direction in the material model

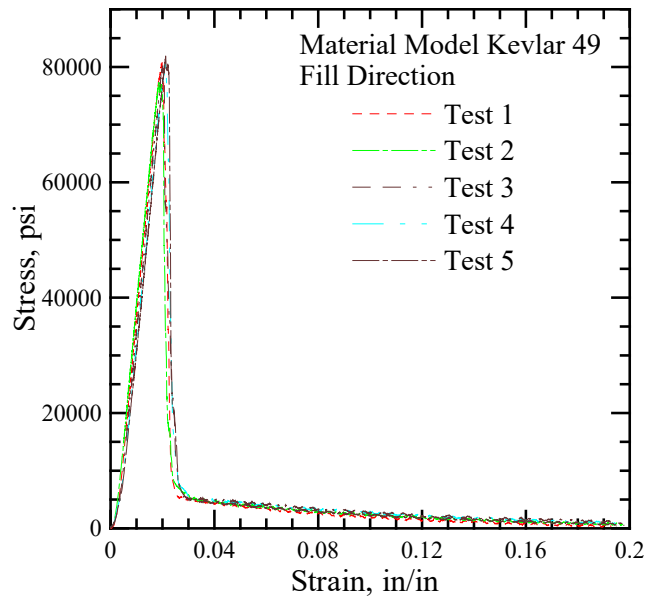
Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	78296	4699006	1262	0.0193
2	82202	4738753	1326	0.0202
3	84543	4824410	1383	0.0206
4	84848	4993236	1297	0.0189
5	79767	4842756	1347	0.0211
Average	81931	4819632	1323	0.0200
Std. Dev.	2886	113785	47	0.0009

8"x1" specimens in (a) warp and (b) fill directions in the material model



Summary of 8"x2" specimens in warp direction in the material model

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	66827	4505431	1330	0.0231
2	68969	4766734	1169	0.0208
3	72260	4791127	1222	0.0239
4	68844	4637988	1438	0.0224
5	71555	4697806	1361	0.0213
Average	69691	4679817	1304	0.0223
Std. Dev.	2209	114486	108	0.0013

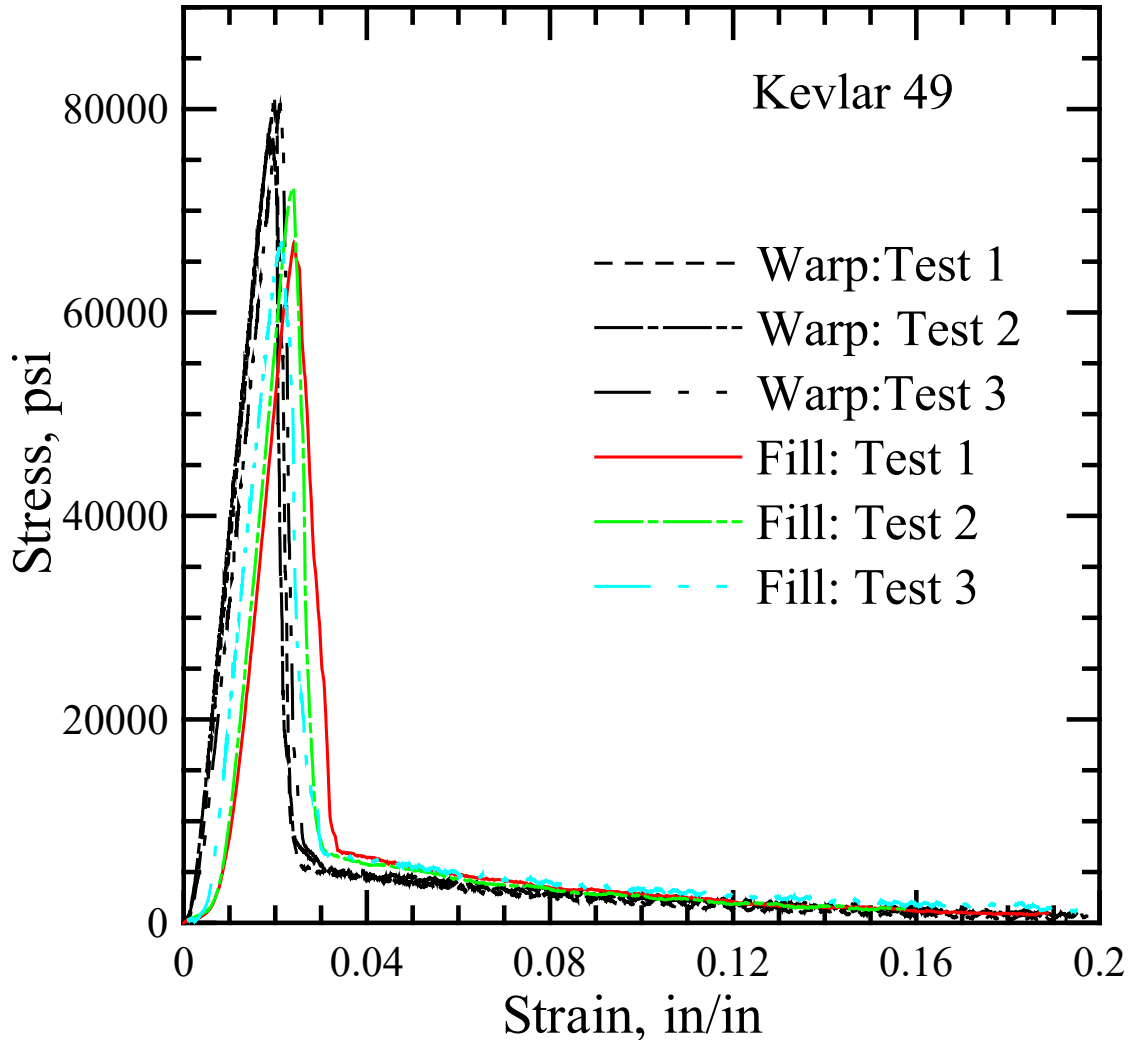


Summary of 8"x2" specimens in fill direction in the material model

Test No	Strength, psi	Modulus, psi	Toughness, psi	Ultimate strain, in/in
1	80991	4818850	1271	0.0198
2	77316	4743059	1332	0.0186
3	80613	4716762	1401	0.0201
4	80549	4533814	1394	0.0210
5	81928	4568982	1355	0.0212
Average	80280	4676293	1351	0.0201
Std. Dev.	1746	120659	53	0.0010

8"x2" specimens in (a) warp and (b) fill directions in the material model

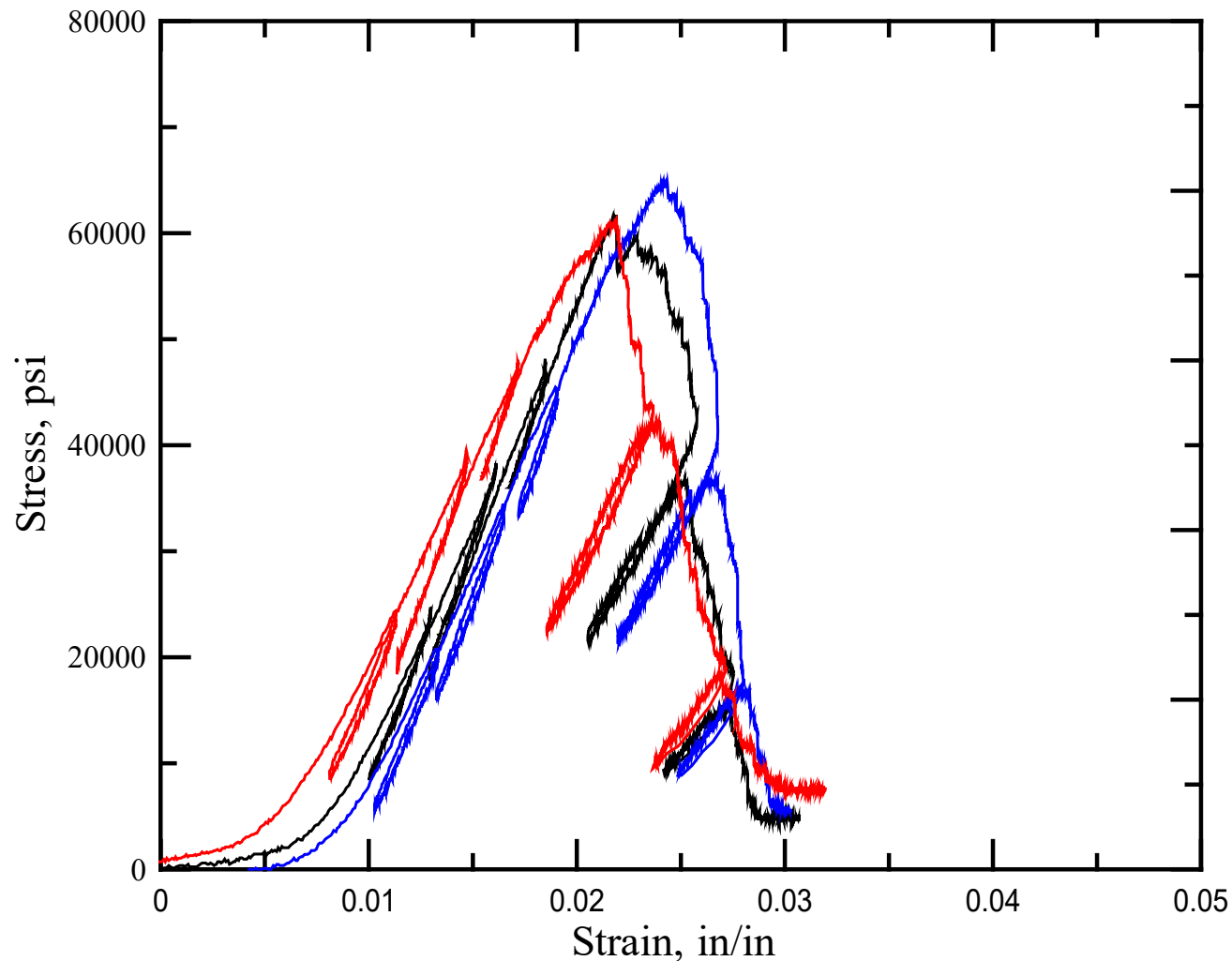
Comparison of Warp and Fill Direction of 8"x2" Specimens



Cyclic Loading Test Results

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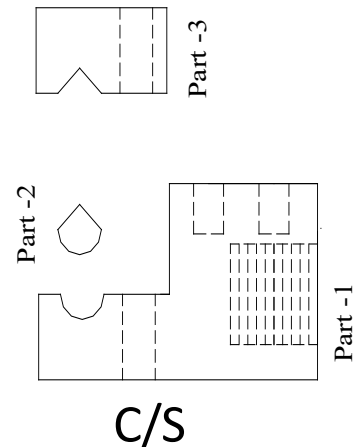
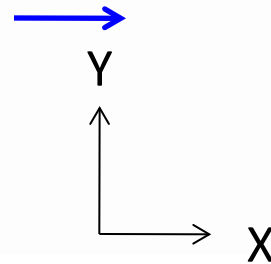
Cyclic testing showed that the fabric unloaded and reloaded along approximately the same path but at a stiffness greater than the elastic stiffness



Biaxial Testing: Poisson's Ratio

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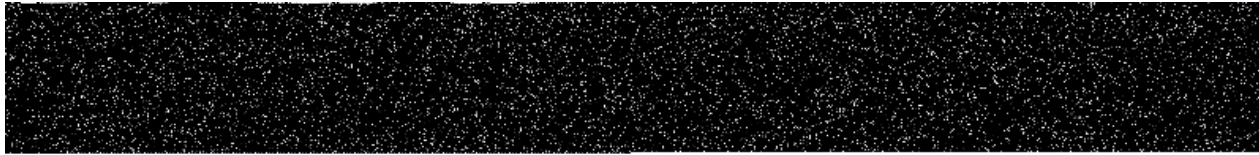
Test Setup



Specimens with Corner Cut-off

Specimen in Fixture

Biaxial Test Video



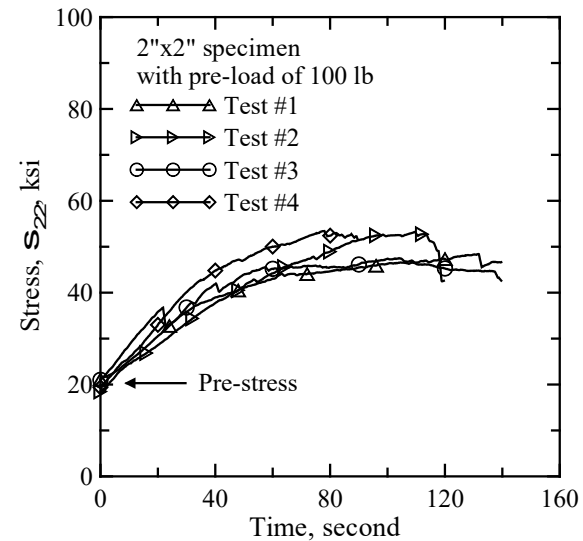
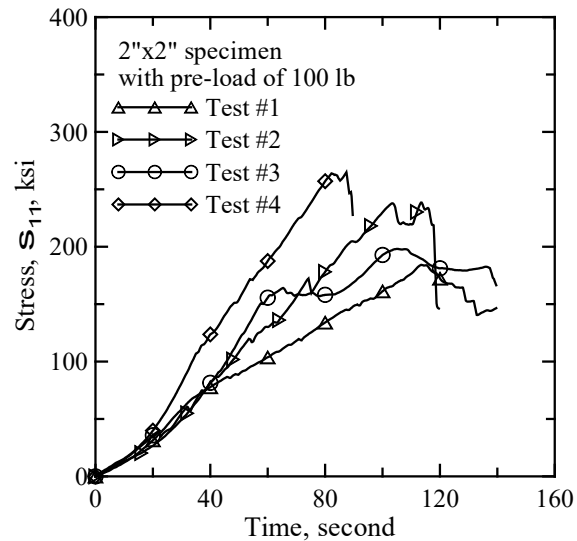
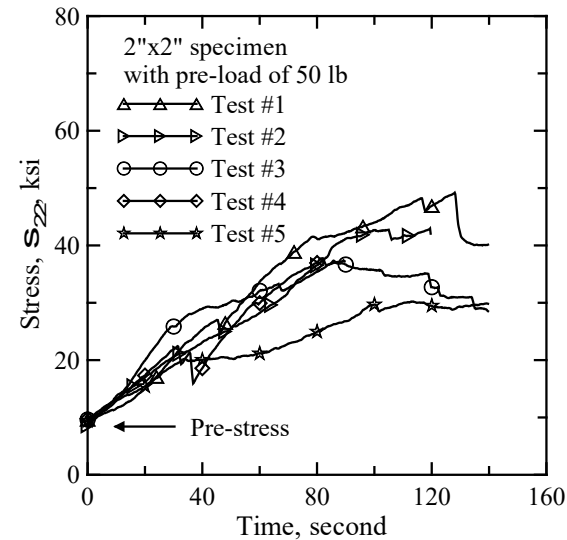
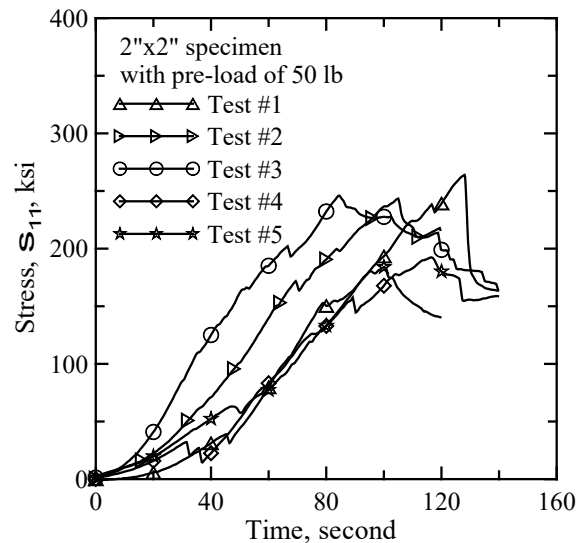
Data Analysis of Biaxial Test

In plane stress case for orthotropic condition

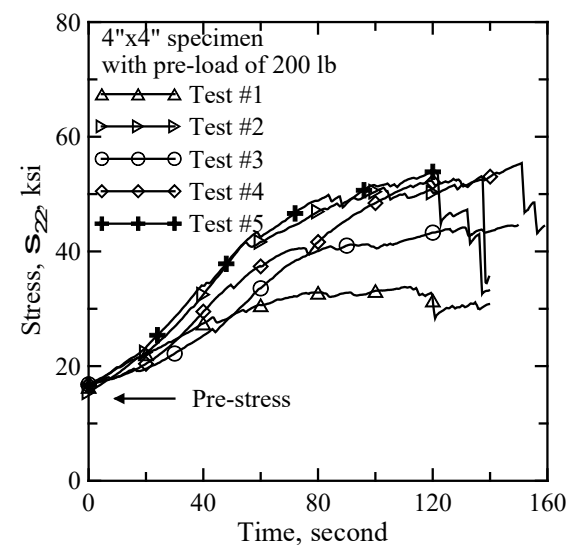
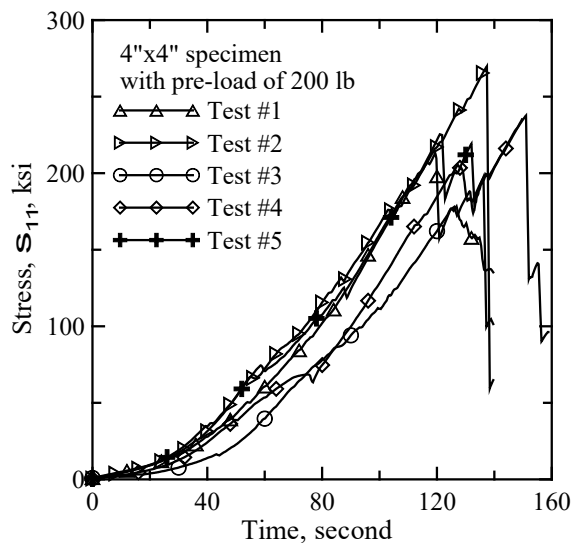
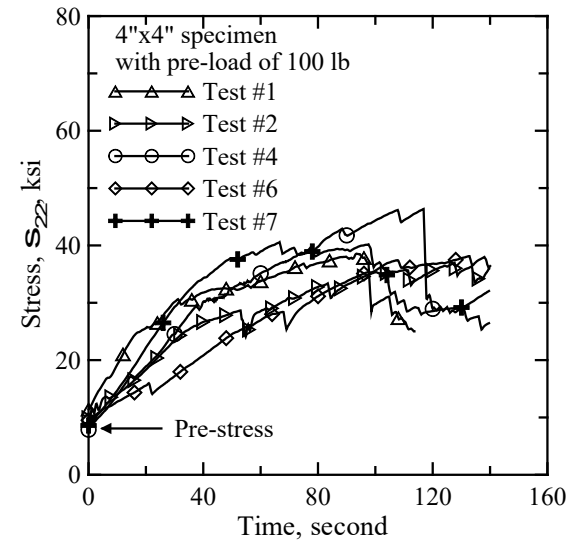
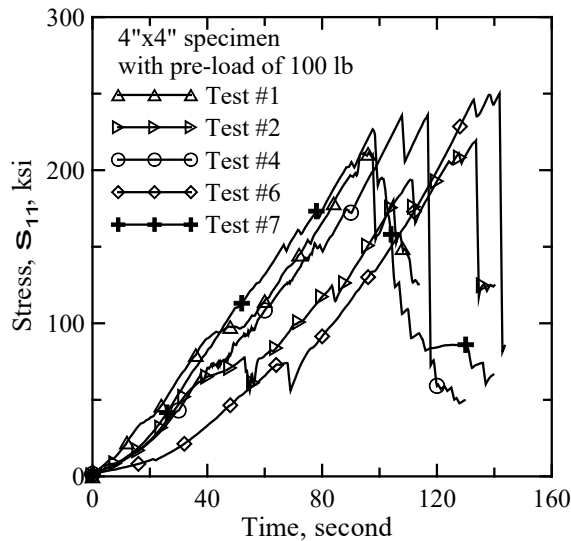
$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{31} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} \frac{E_{11}}{1-\nu_{12}\nu_{21}} & \frac{\nu_{21}E_{11}}{1-\nu_{12}\nu_{21}} & 0 & 0 & 0 & 0 \\ \frac{\nu_{12}E_{22}}{1-\nu_{12}\nu_{21}} & \frac{E_{22}}{1-\nu_{12}\nu_{21}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2G_{12} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{23} \\ \varepsilon_{31} \\ \varepsilon_{12} \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \begin{cases} \sigma_{11} = \frac{E_{11}}{1-\nu_{12}\nu_{21}} \varepsilon_{11} + \frac{\nu_{21}E_{11}}{1-\nu_{12}\nu_{21}} \varepsilon_{22} \\ \sigma_{22} = \frac{\nu_{12}E_{22}}{1-\nu_{12}\nu_{21}} \varepsilon_{11} + \frac{E_{22}}{1-\nu_{12}\nu_{21}} \varepsilon_{22} \end{cases} & \Rightarrow \begin{cases} \nu_{12} = \frac{(\sigma_{22} - E_{22}\varepsilon_{22})E_{11}}{\sigma_{11}E_{22}} \\ \nu_{21} = \frac{(\sigma_{22} - E_{22}\varepsilon_{22})}{\sigma_{11}} \end{cases} \end{aligned}$$

Response of 2"x2" Specimens

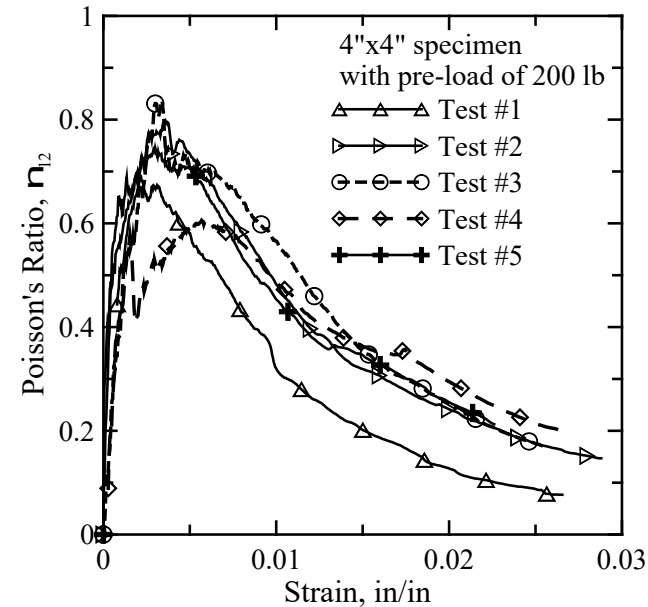
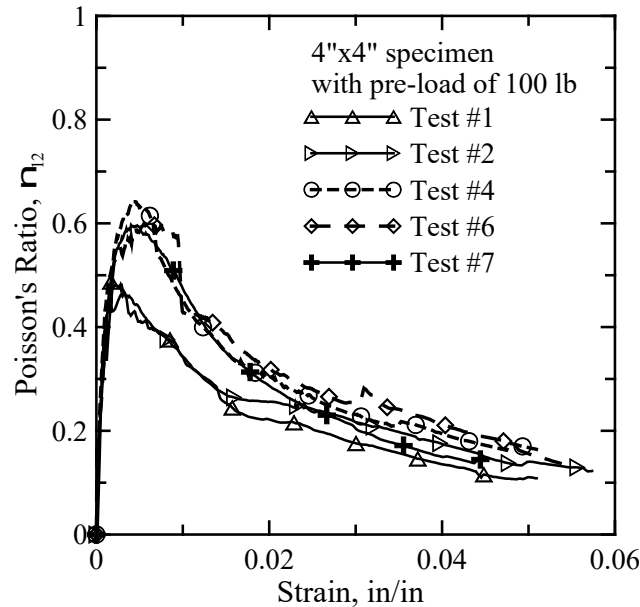
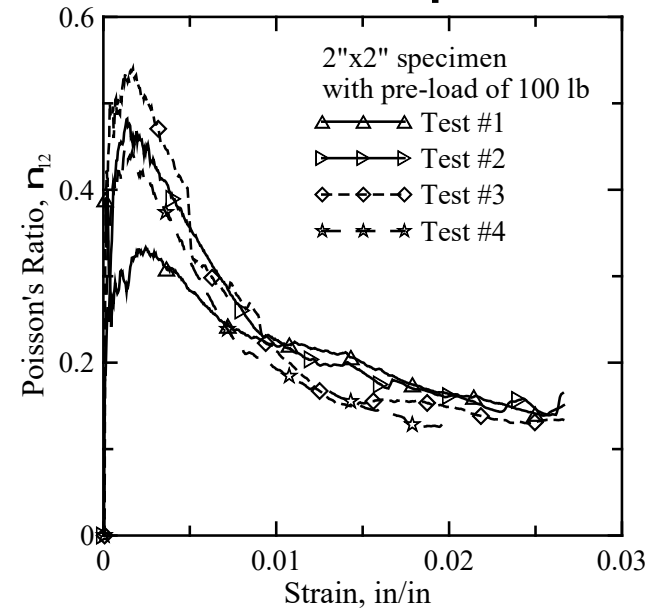
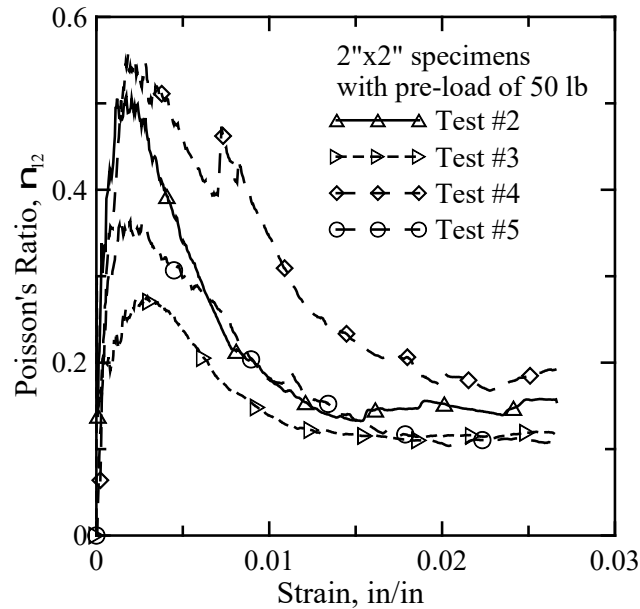


Response of 4"x4" Specimens

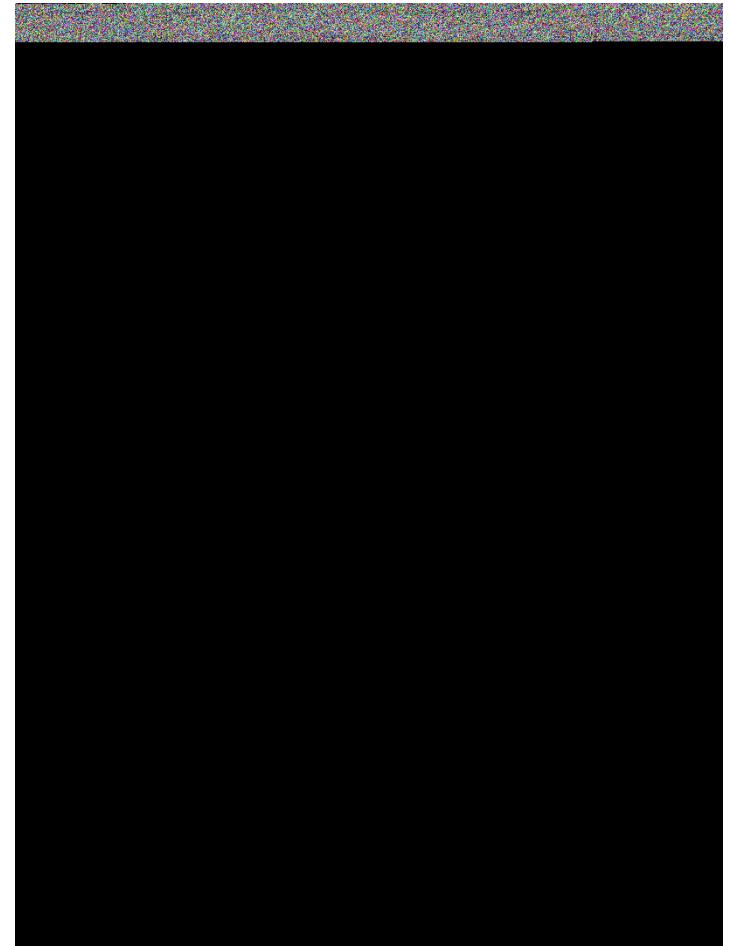
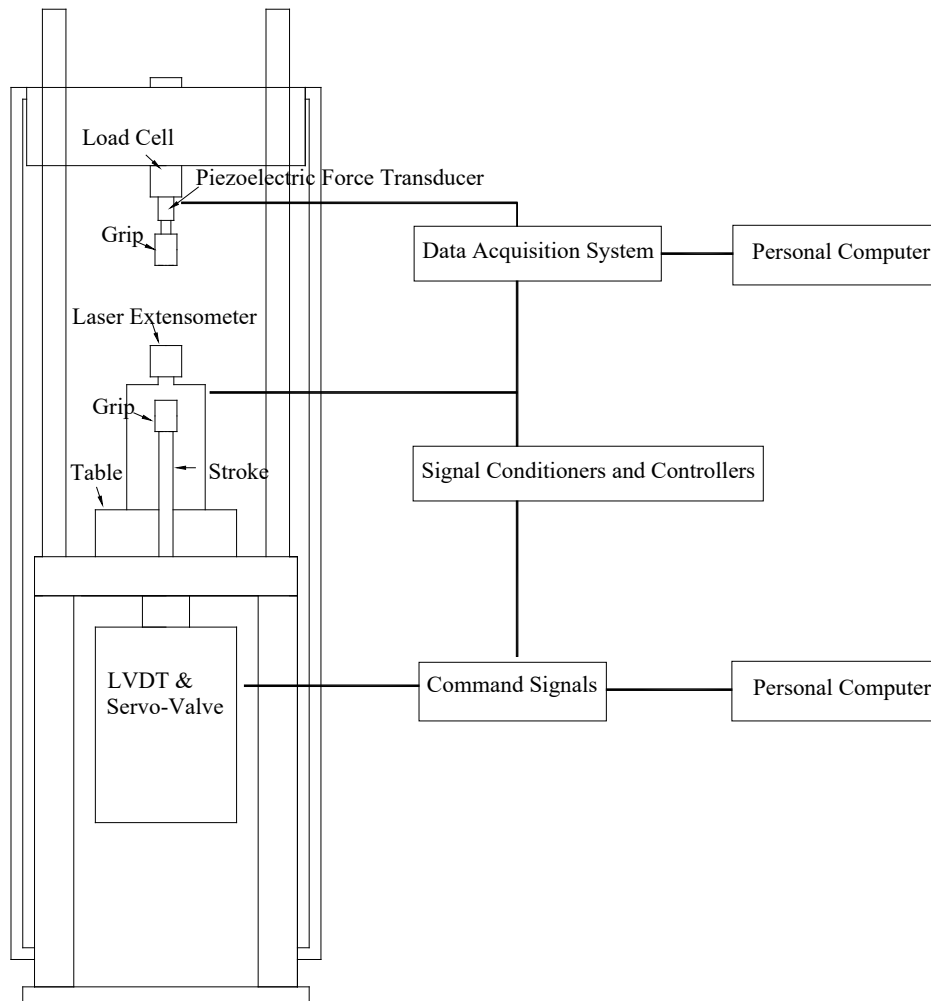


Poisson's Ratio for 2"x2" and 4"x4" Specimens

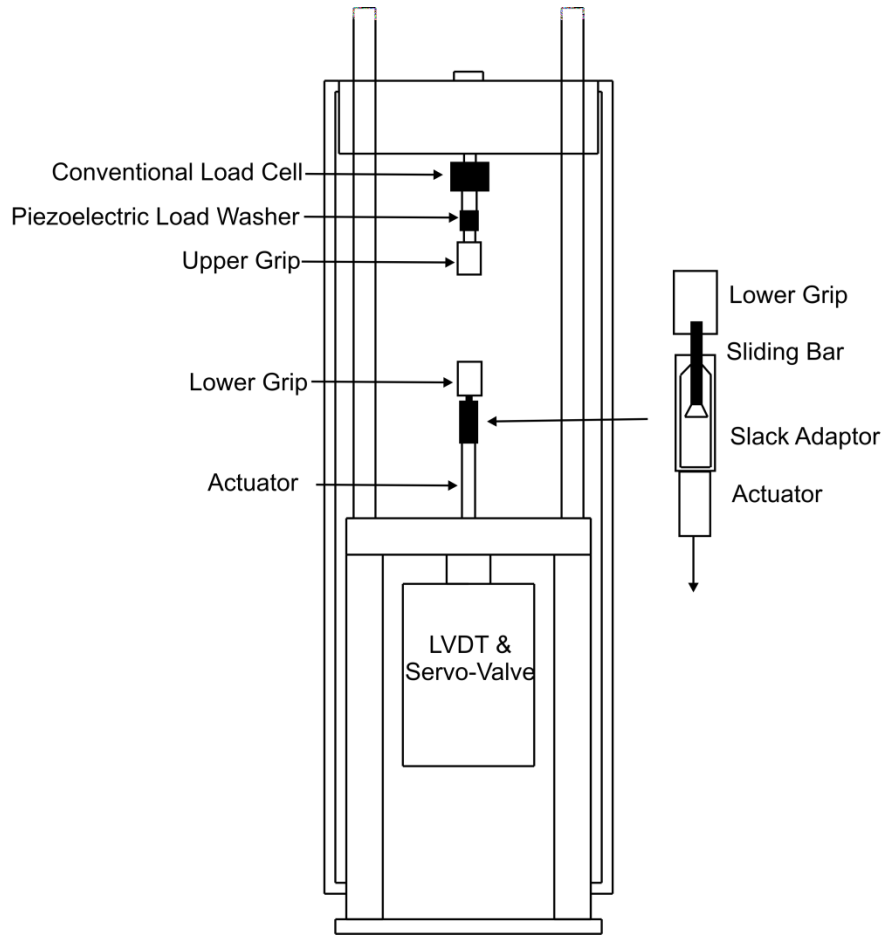
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High Speed Testing System- Apparatus



Vibration Characteristics of the²⁰ Testing System

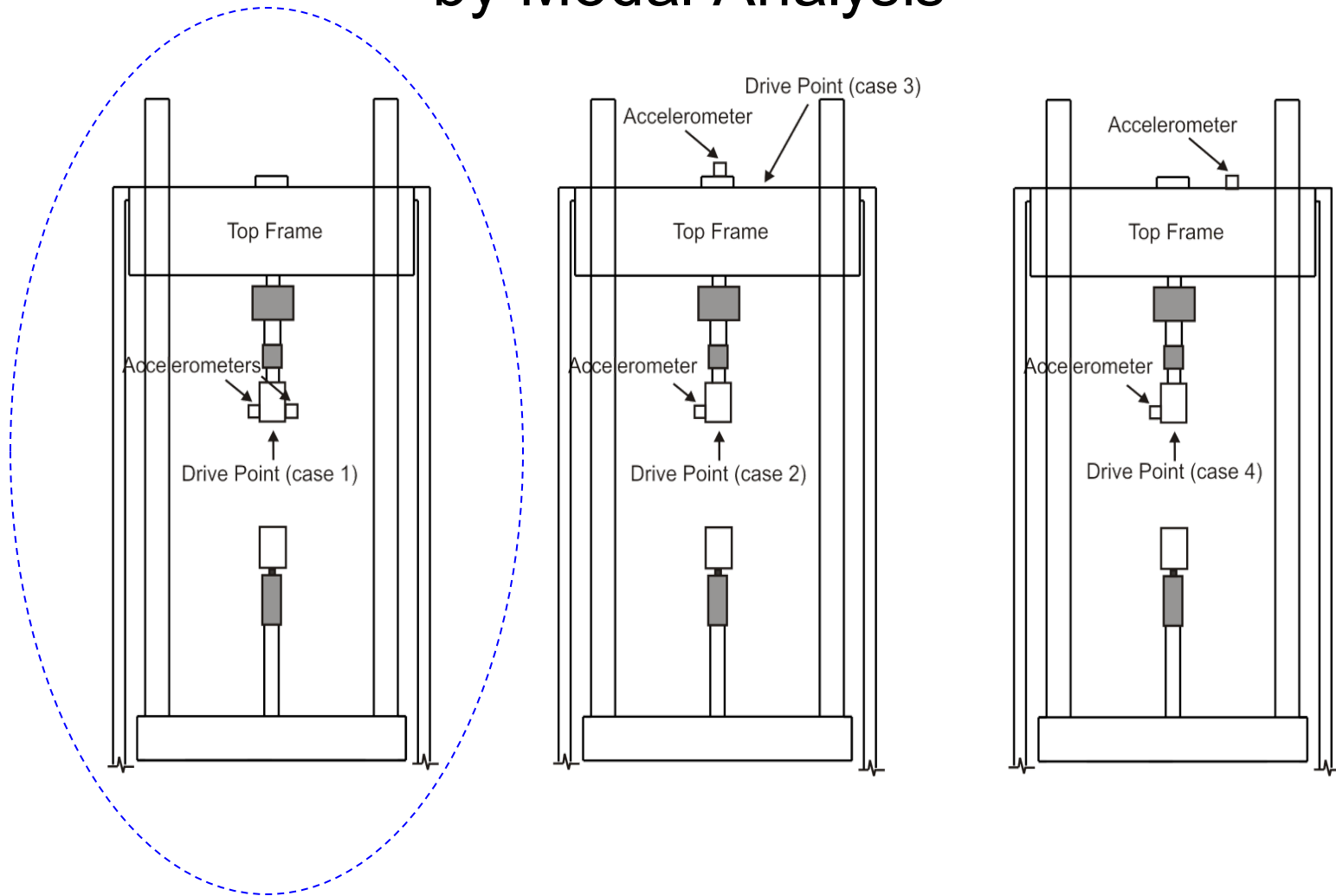


- System Ringing: an important issue in dynamic material testing using a servo-hydraulic tensile testing machine.
- The vibration response can be analyzed using a single degree-of-freedom (SDOF) spring-mass-damper. The response is given by:

$$x = \frac{V}{\omega_n \sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin \sqrt{1 - \zeta^2} \omega_n t$$

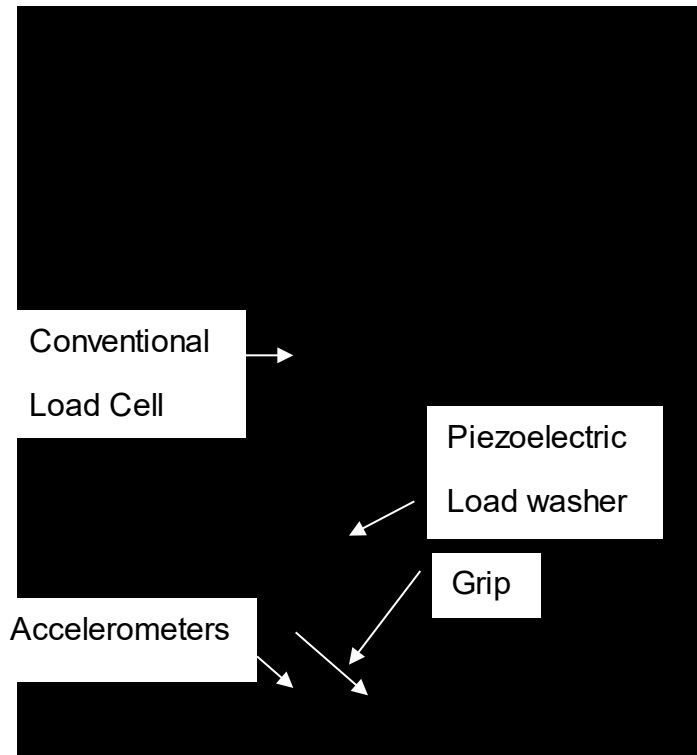
$$\sigma = \frac{EV}{L \omega_n \sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin \sqrt{1 - \zeta^2} \omega_n t$$

Vibration Characteristics of High Speed Machine²¹ by Modal Analysis

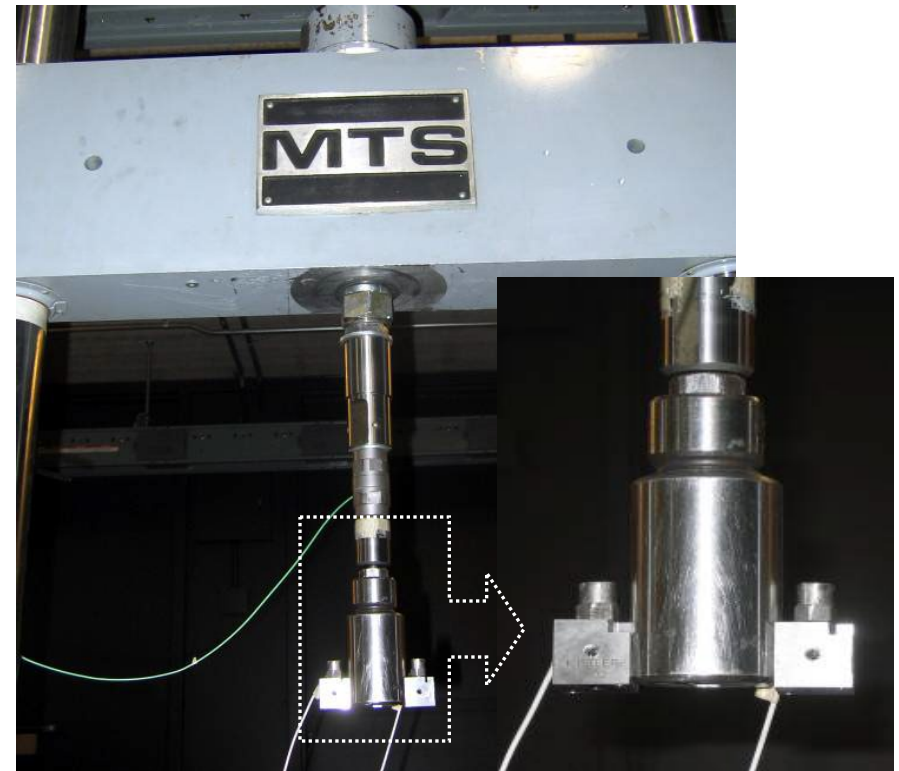


Schematics of Test Cases

Modal Analysis Setup of Case 1

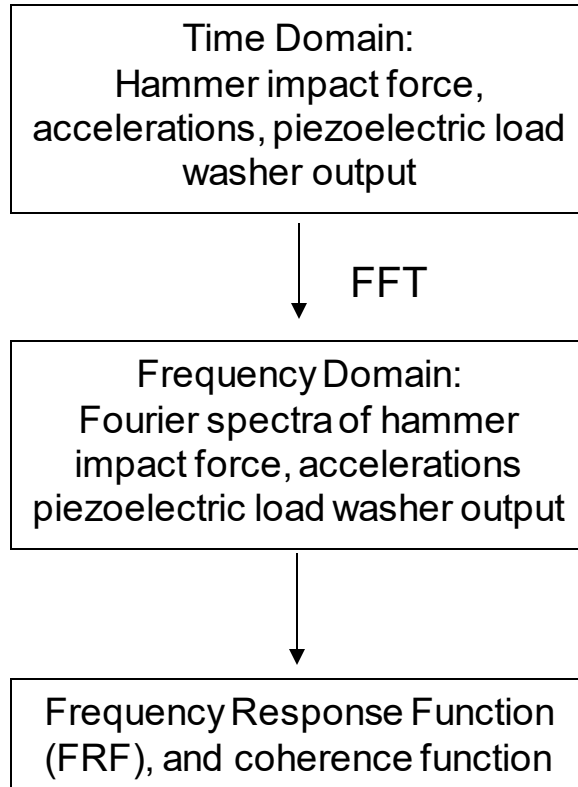


Original Configuration



New Configuration

Data Processing of Modal Analysis



Flow Chart of Data Processing

Cross power spectrum:

$$G_{xy}(f) = S_x(f)S_y^*(f)$$

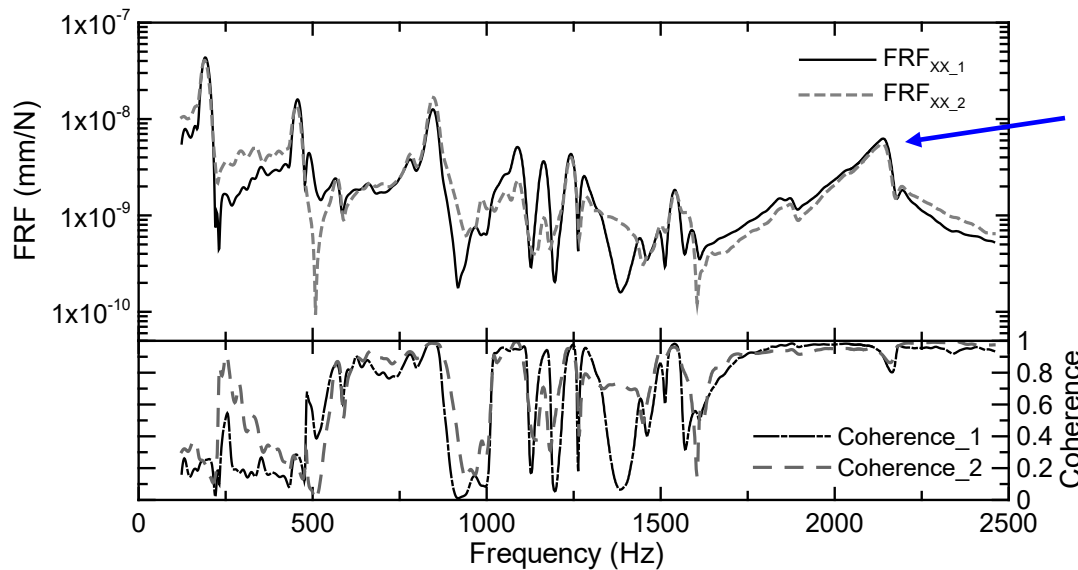
where $S_x(f)$ is the Fourier transform of the input signal (hammer impact force), $S_y(f)$ is the Fourier transform of the output signal (acceleration), and $*$ indicates the complex conjugate of the function.

Frequency Response Function:

$$H(f) = \frac{\overline{G_{yx}}(f)}{\overline{G_{xx}}(f)}$$

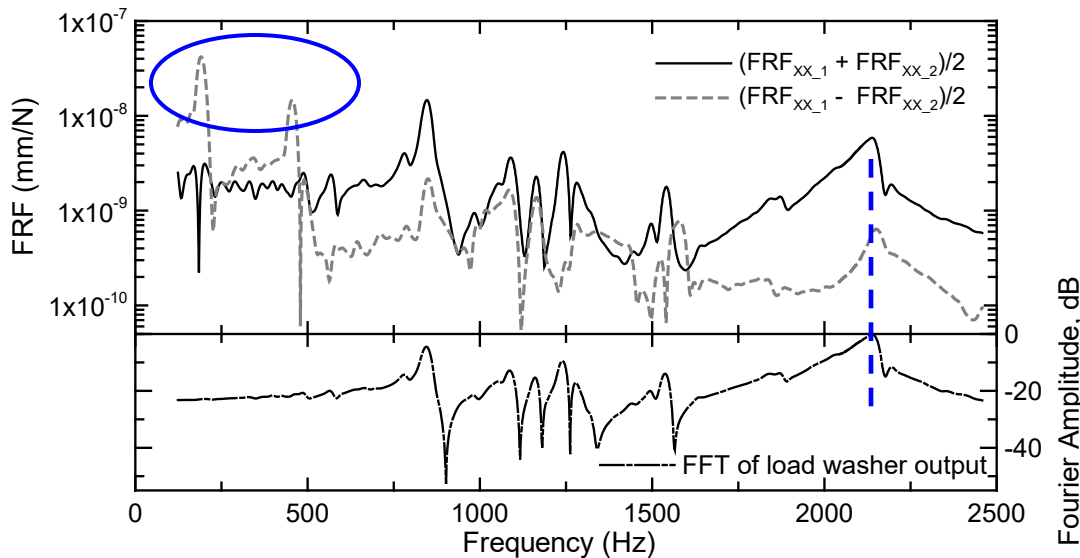
Coherence Function:

$$\gamma_{xy}^2 = \frac{|\overline{G_{xy}}|^2}{\overline{G_{xx}} \overline{G_{yy}}}$$



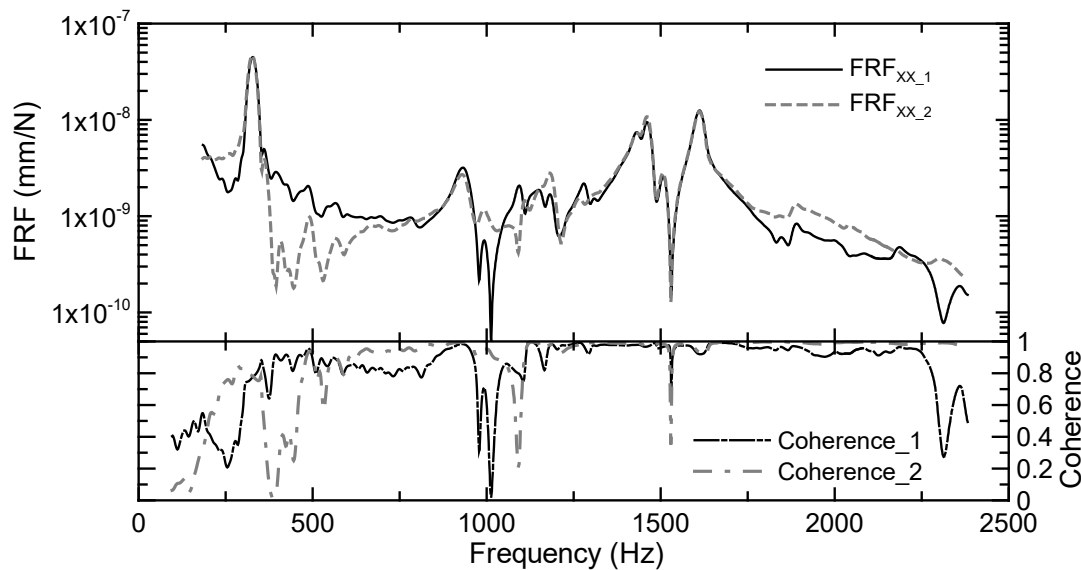
Max. Frequency: 2.2 kHz

FRF's of center rod and coherence functions of Original Configuration



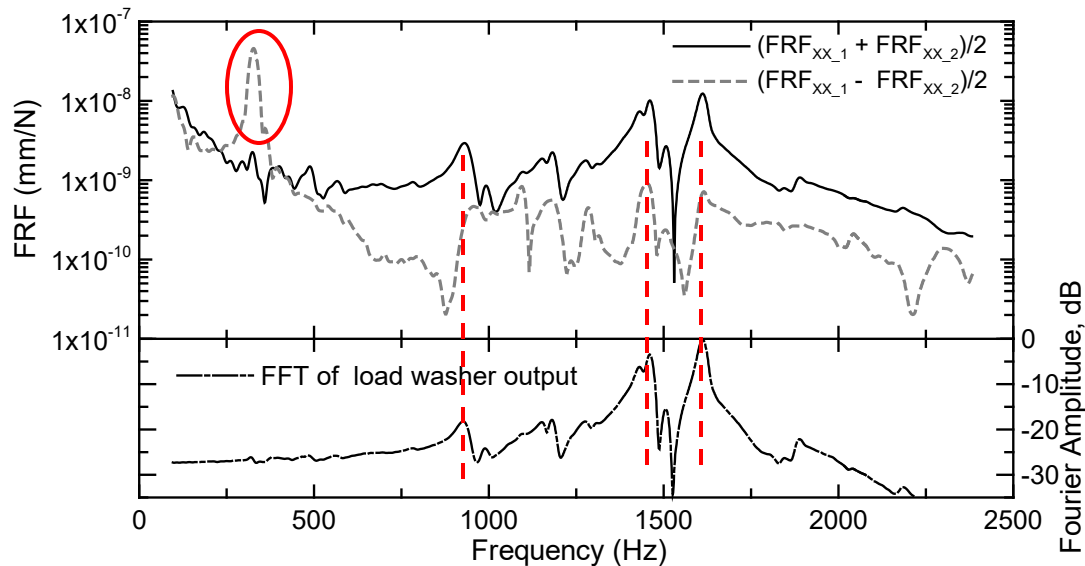
The axial modes of center rod in original configuration are between 0.8 and 2.2 kHz

Summation and difference of FRF's of Original Configuration



There are four modes which are about 0.35, 0.9, 1.4 and 1.6 kHz based on FRF curves.

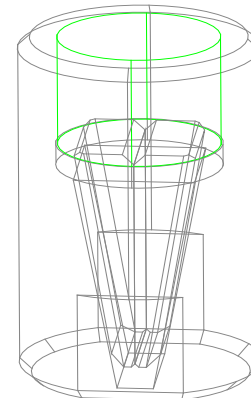
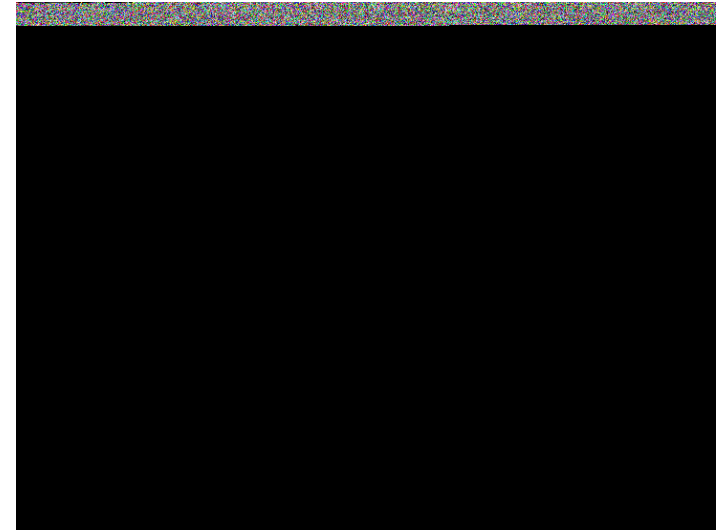
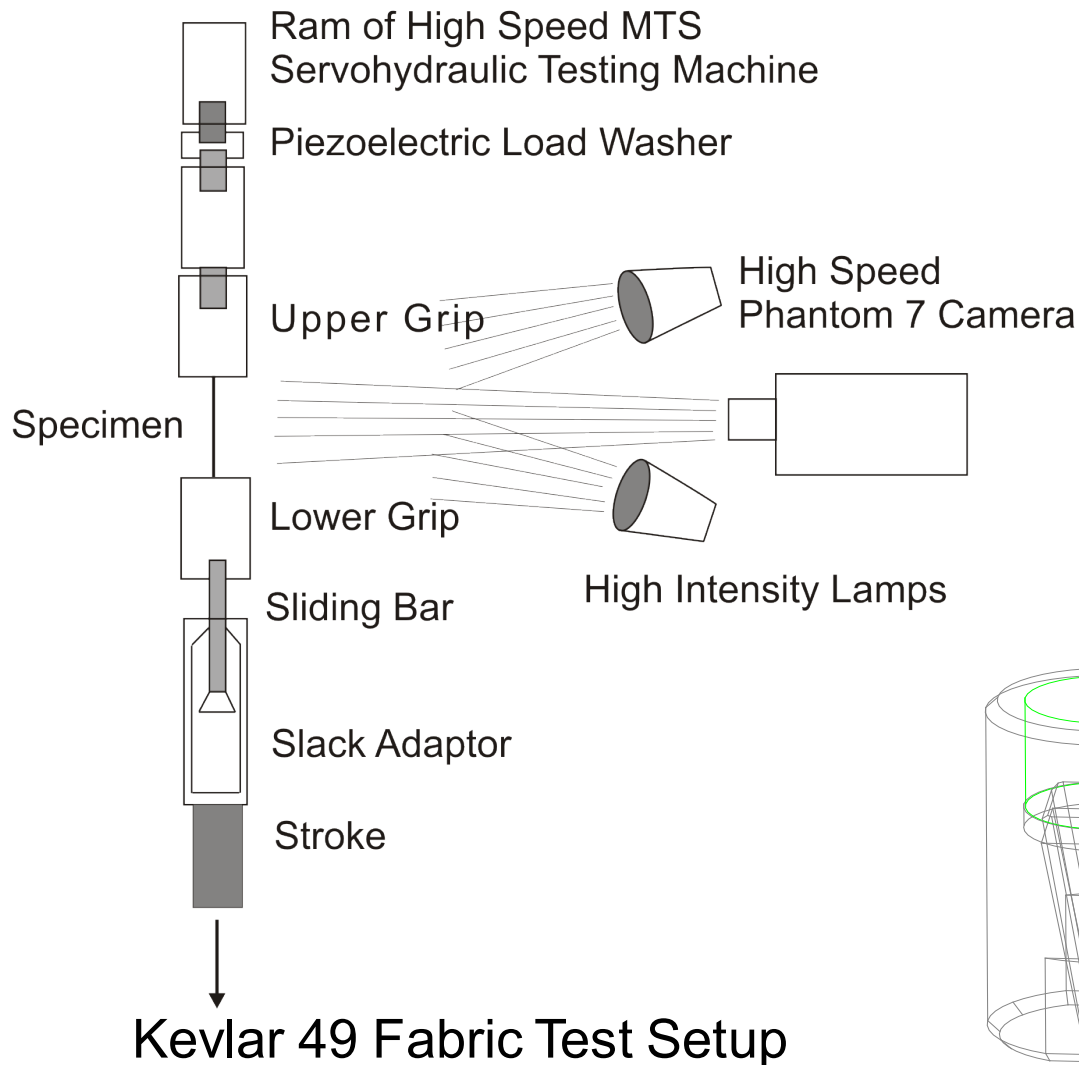
FRF's of center rod and coherence functions of New Configuration



The mode at 0.35 kHz disappears, while the modes at 0.9, 1.4 and 1.6 kHz remain, indicating that the mode at 0.35 kHz is a bending mode of the center rod

Summation and difference of FRF's of New Configuration

Kevlar 49 Fabric Test at High Strain Rate

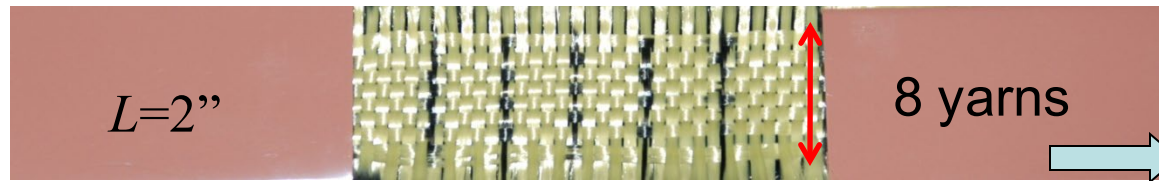


Grip Set

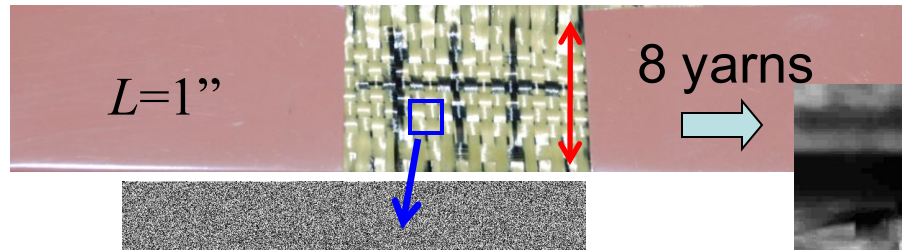
Kevlar 49 Fabric Test at High Strain Rate (continue)

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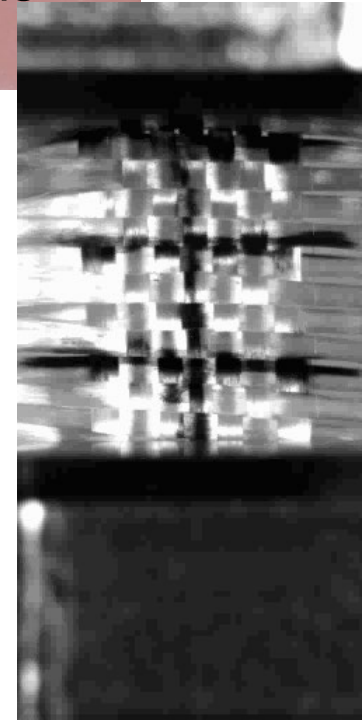
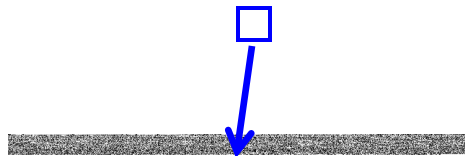
Macro-scale
(Fabric)



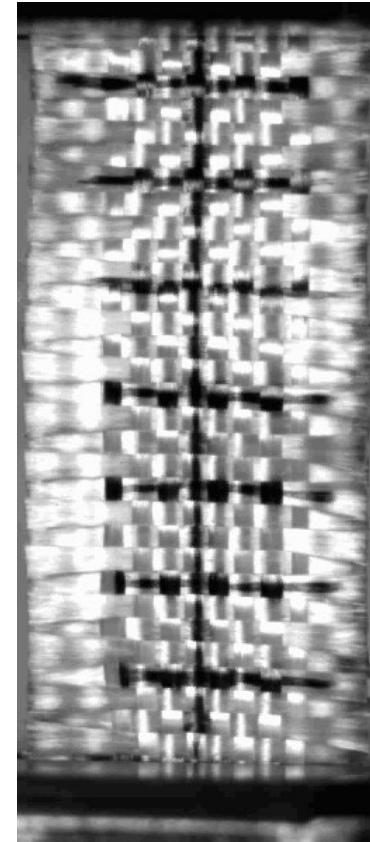
Meso-scale
(Unit Cell)



Micro-scale
(Fiber)

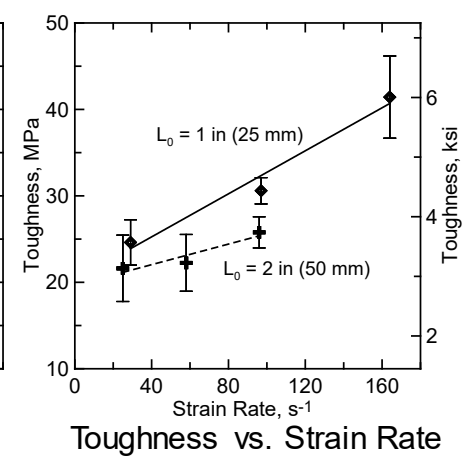
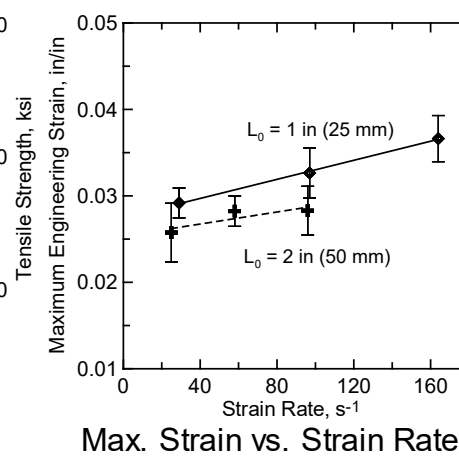
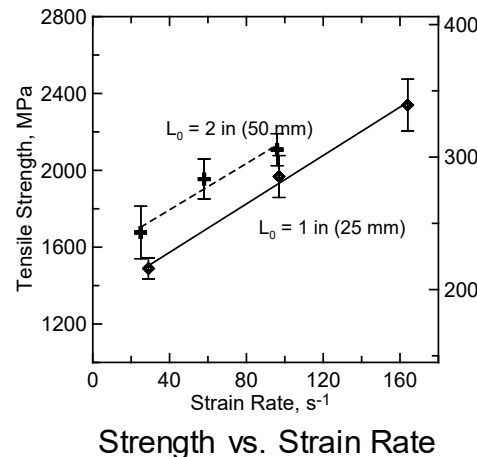
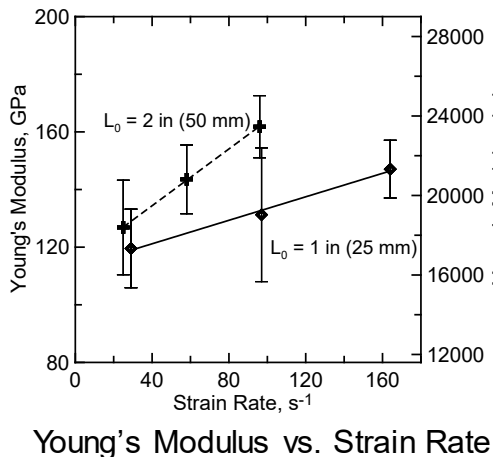
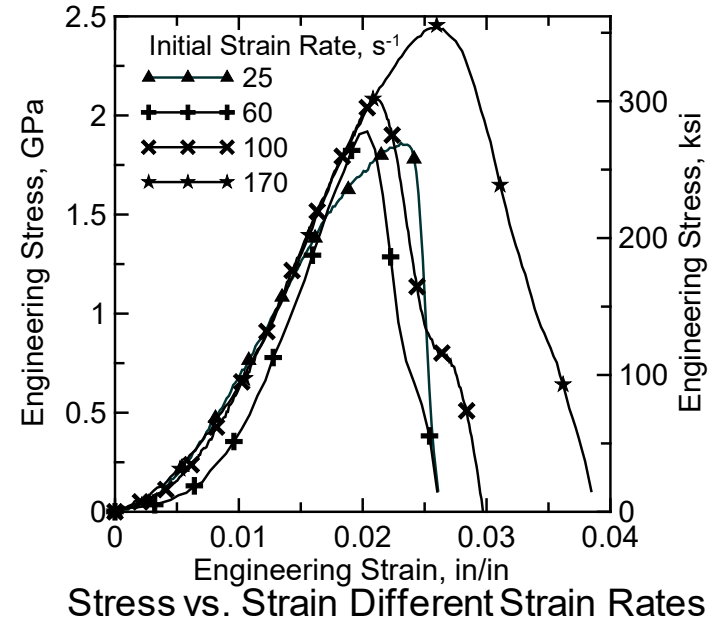
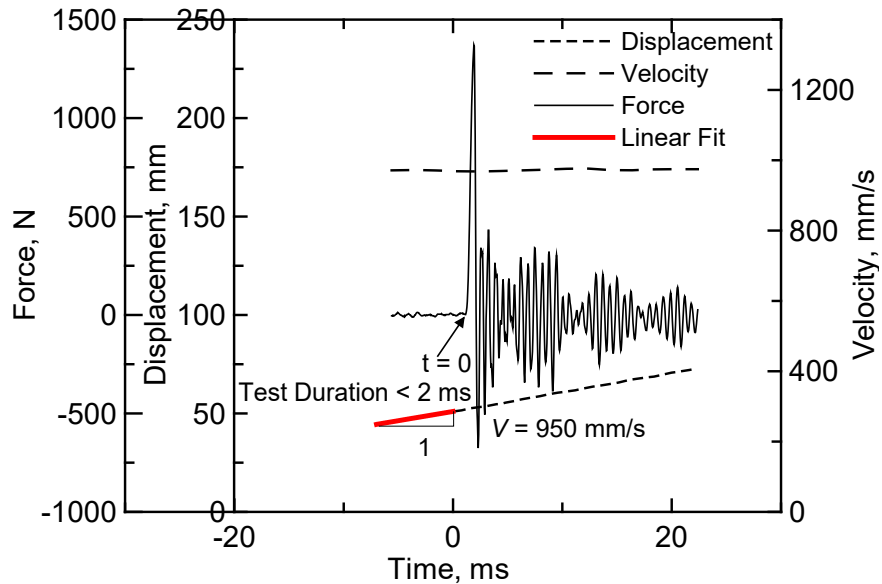


1" Fabric Video 2" Fabric Video

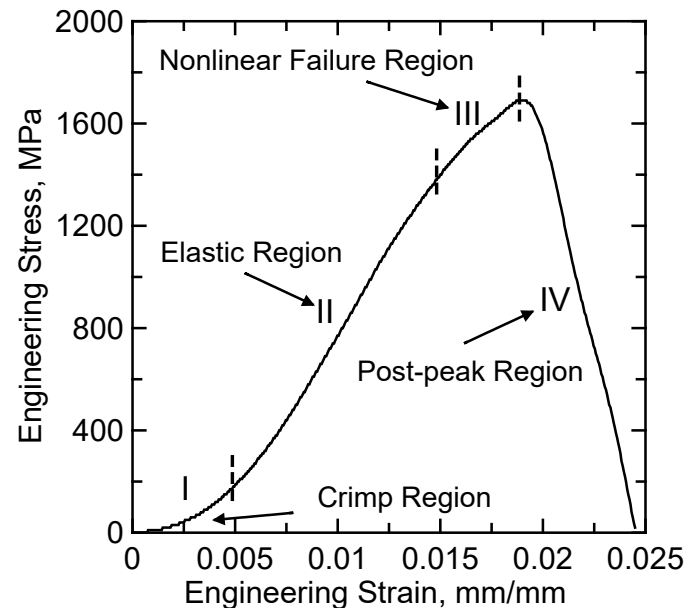
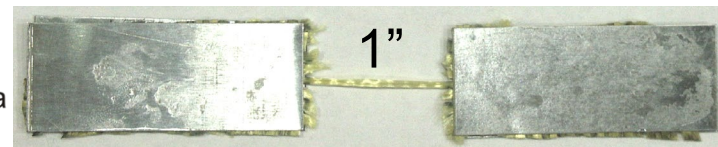
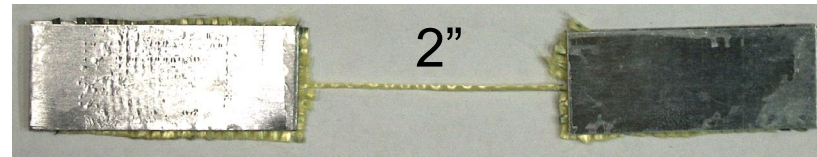
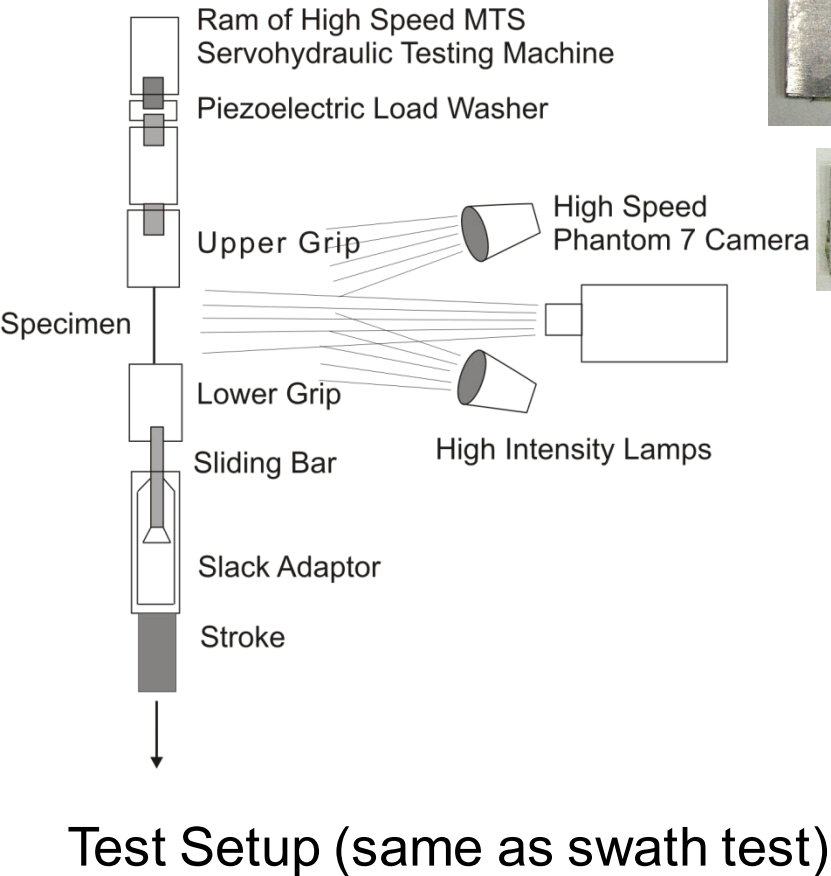


Typical Response and strain rate effect on dynamic material properties

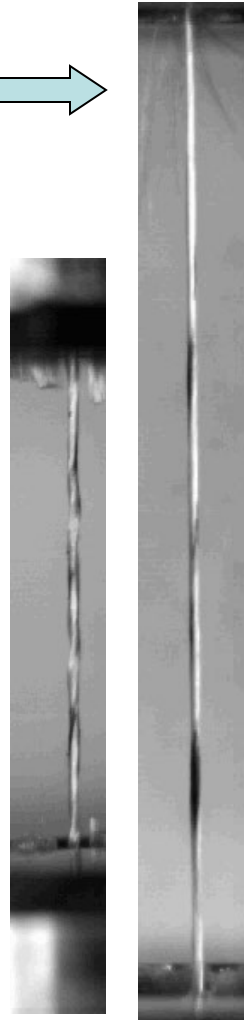
28



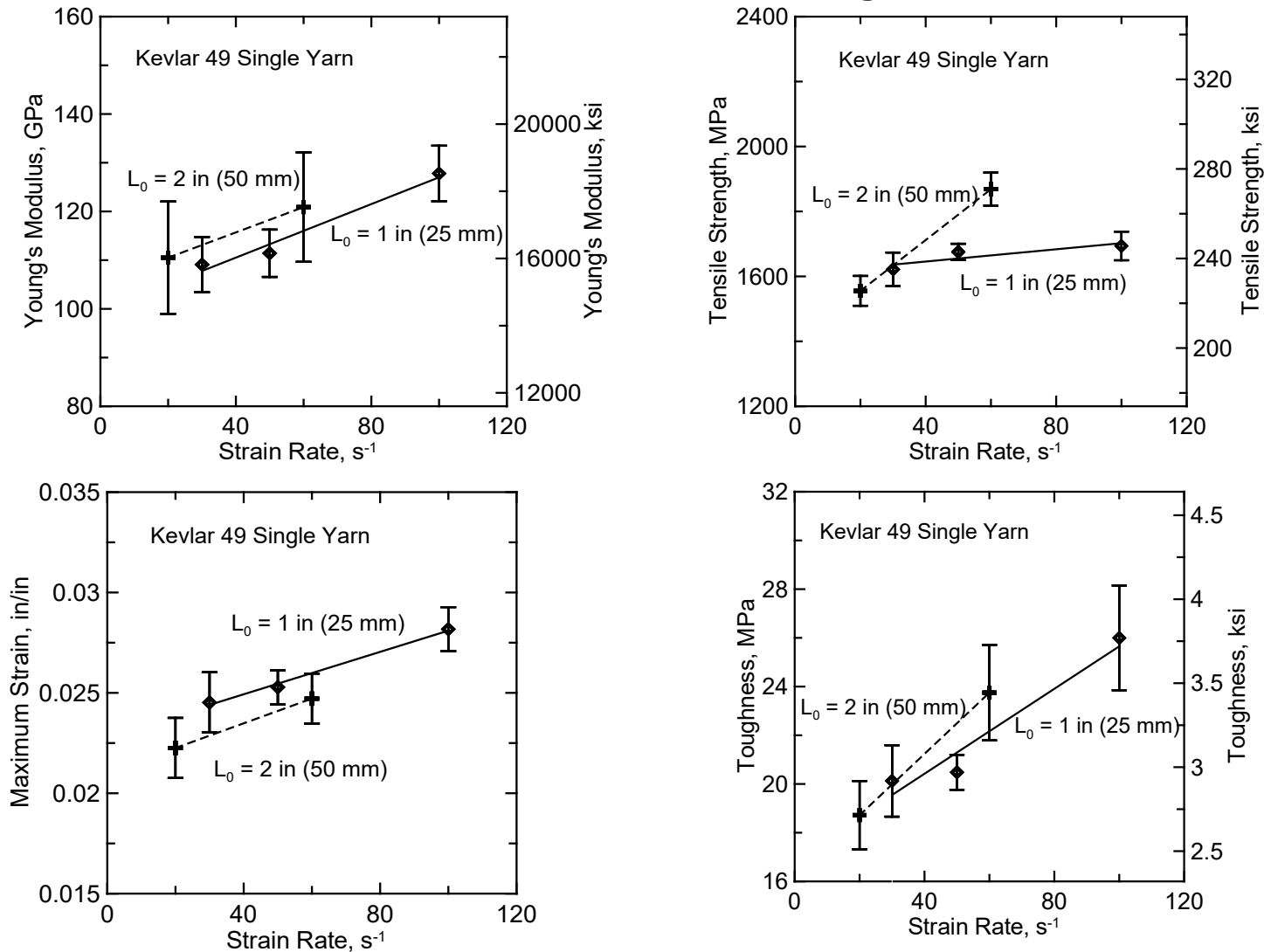
High Strain Rate Test of Kevlar 49 Single Yarn



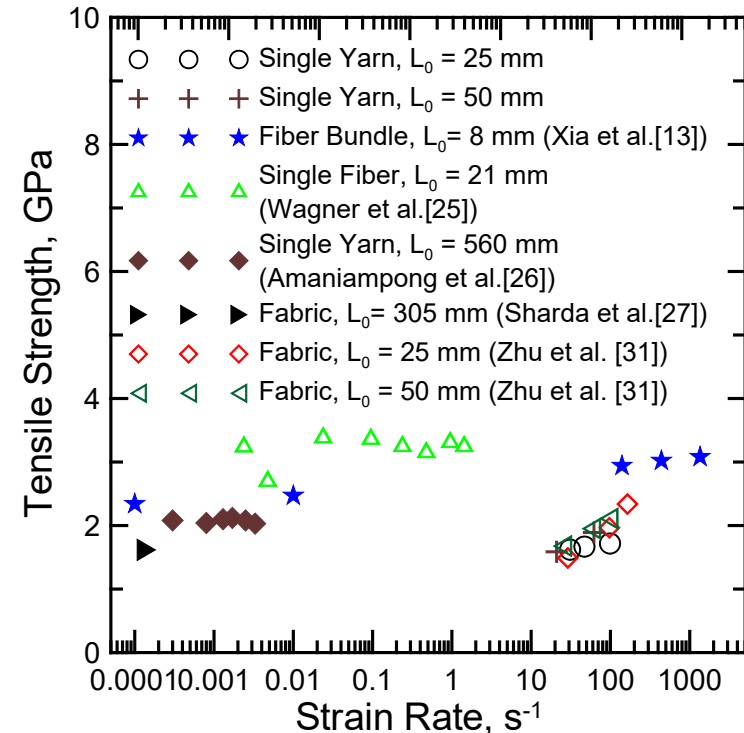
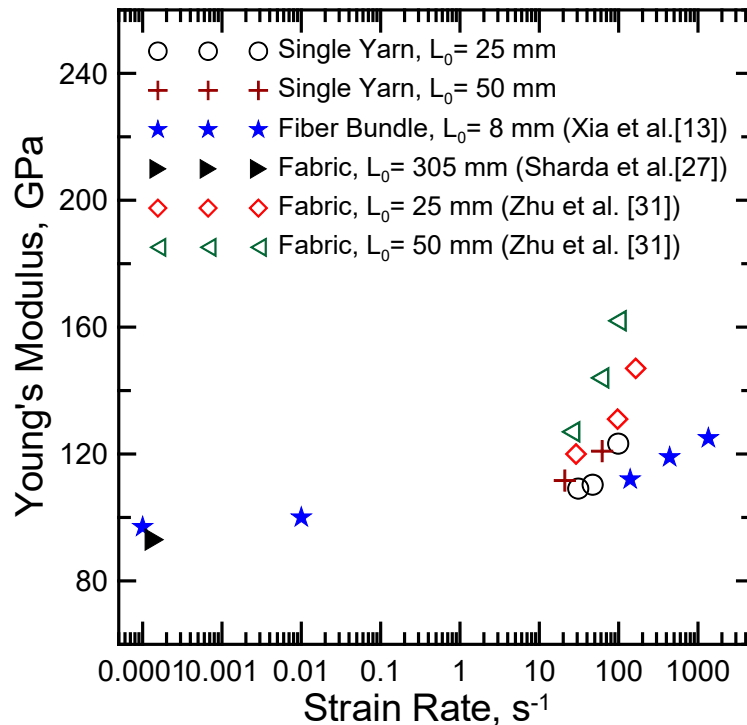
Four regions of the stress-strain curve



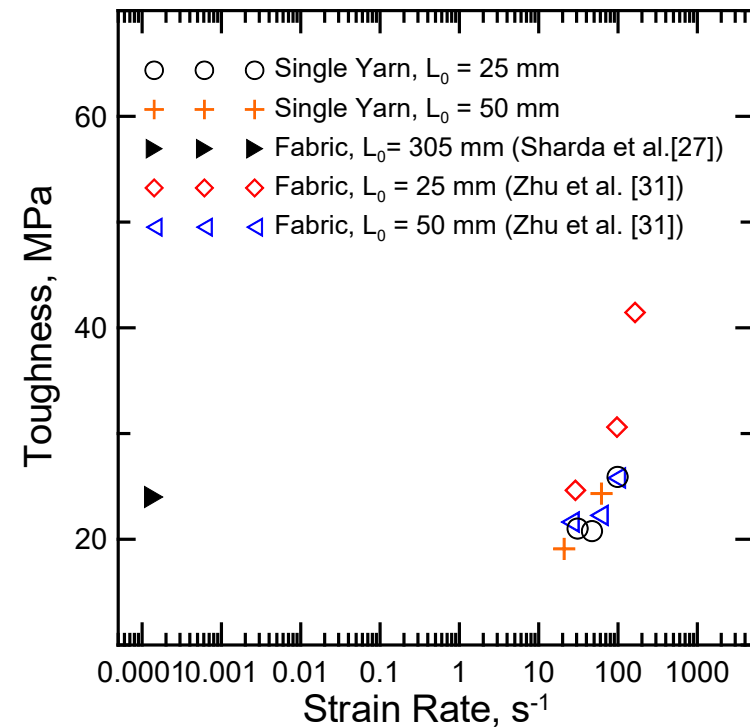
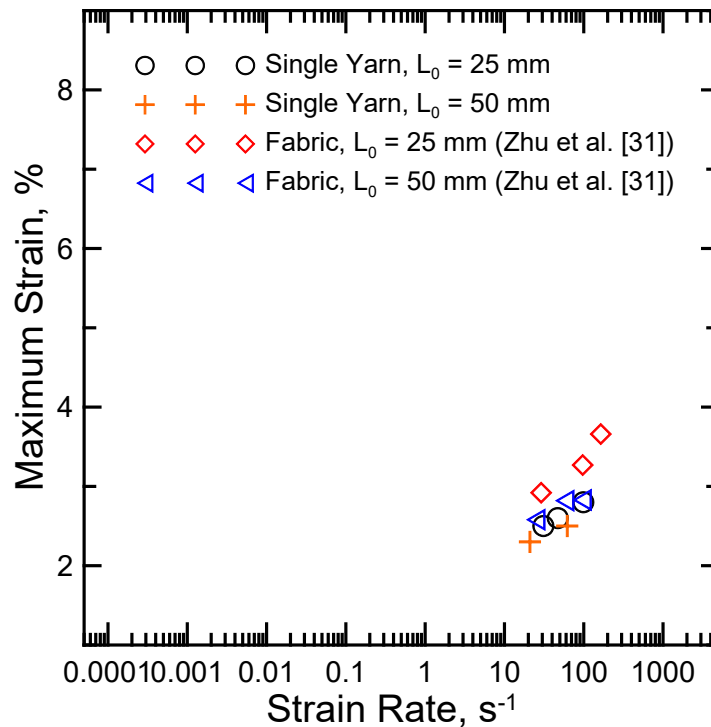
Strain rate effect on dynamic material properties of Kevlar 49 single Yarn



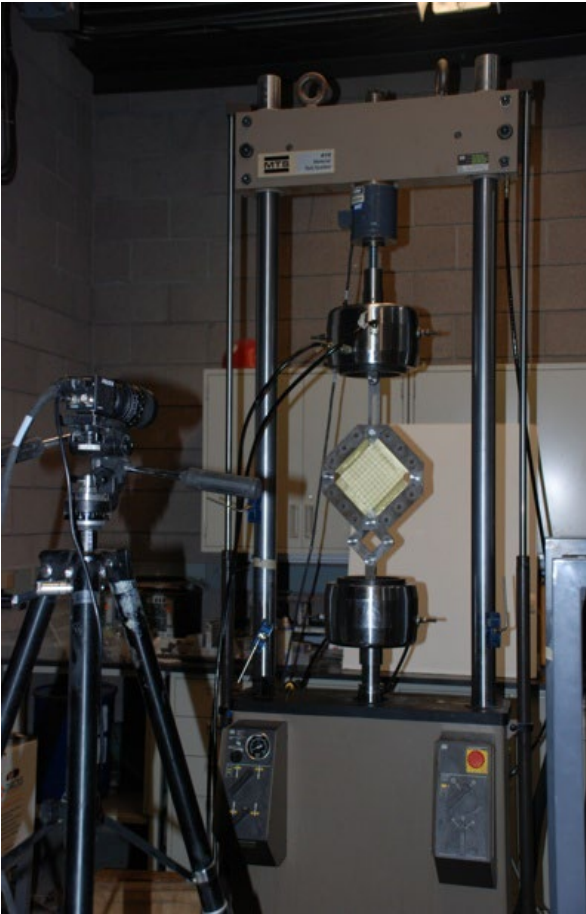
Comparison of Present Study (Fabric and Single Yarn) with Literature Data



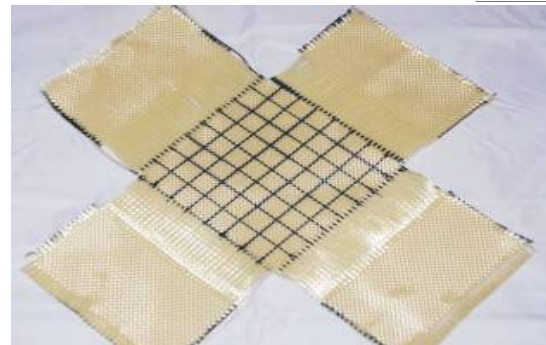
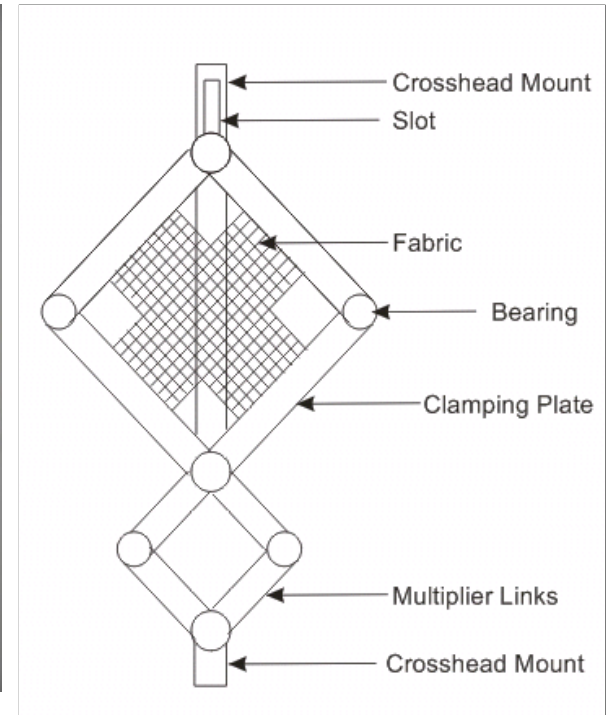
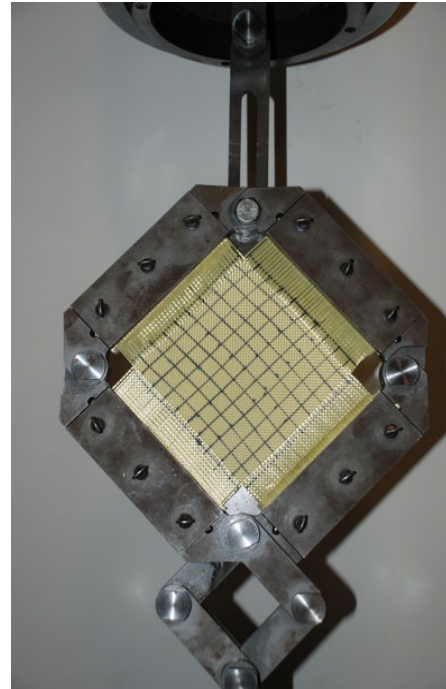
Comparison of Present Study (Fabric and Single Yarn) with Literature Data



Picture Frame Shear Testing



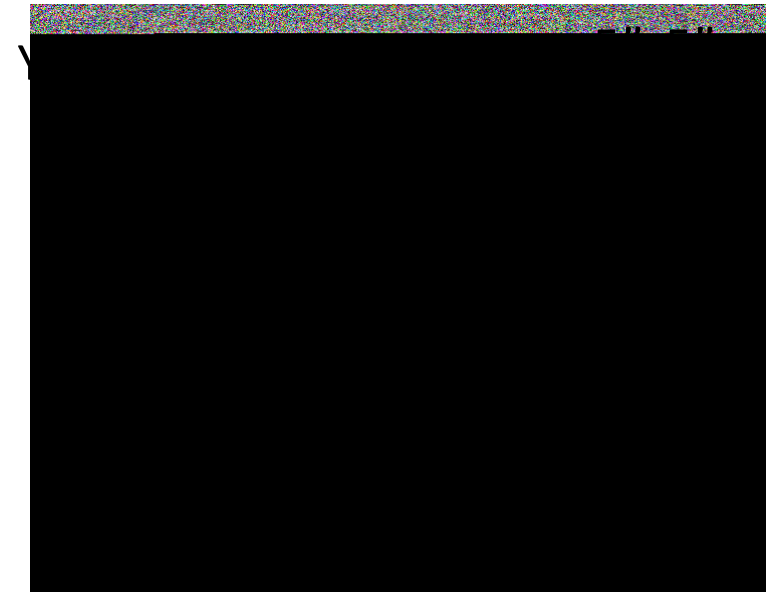
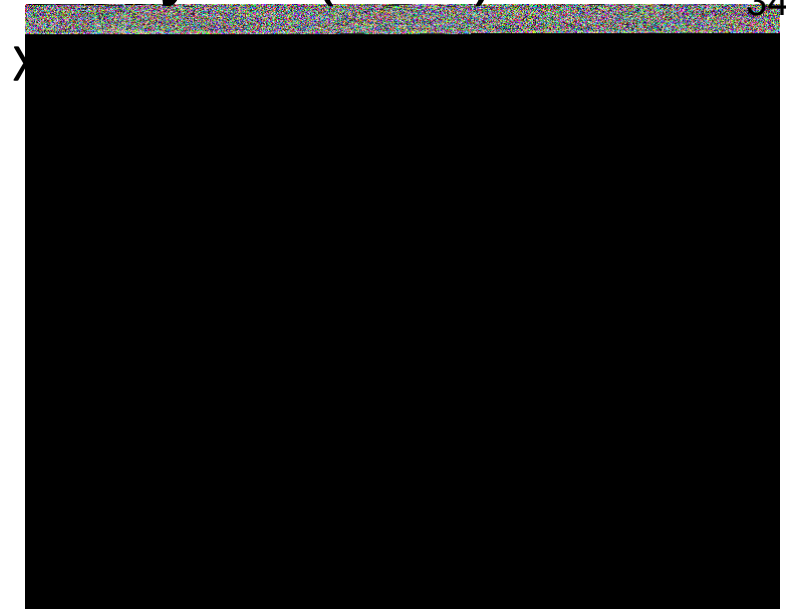
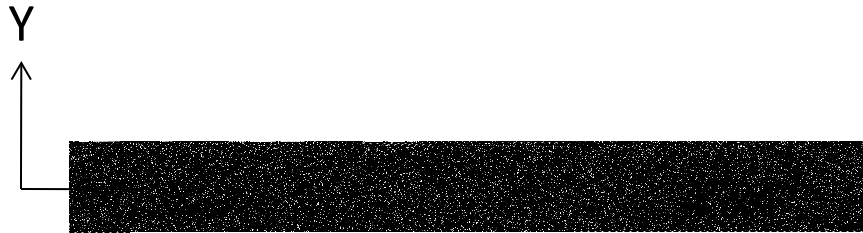
Picture frame test setup

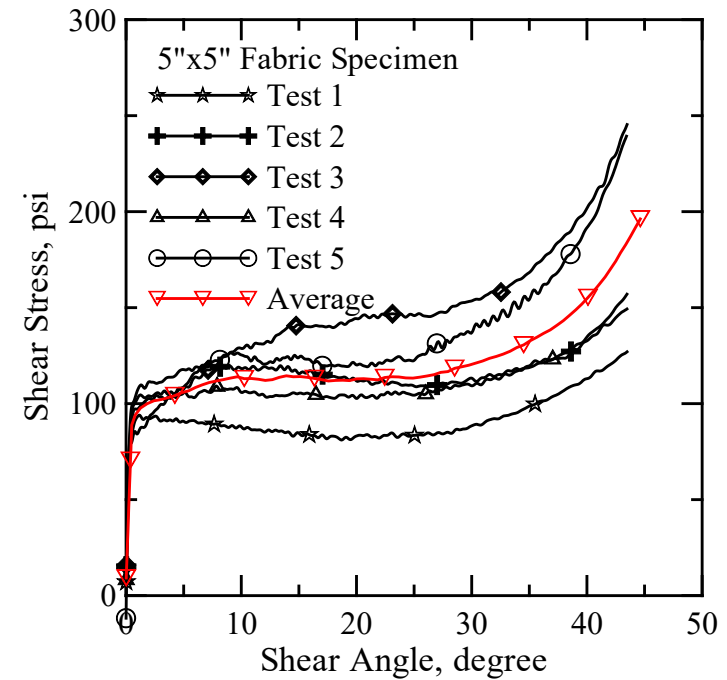
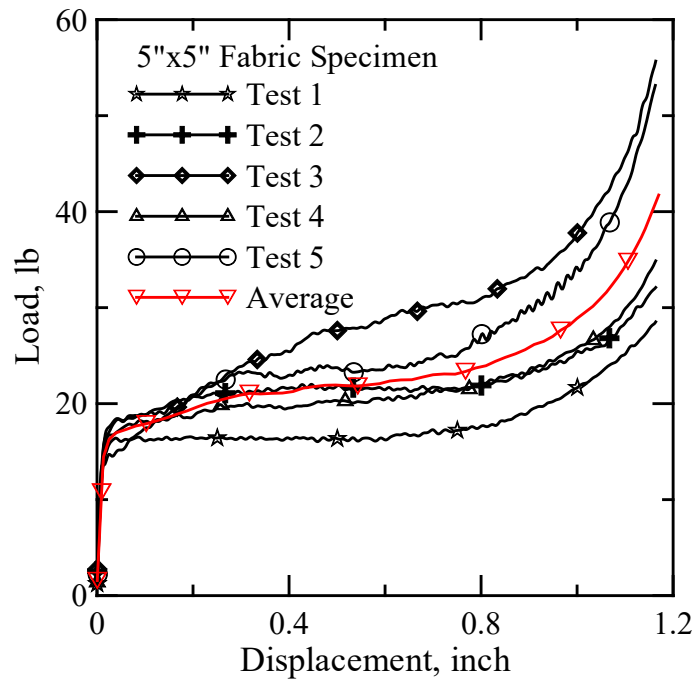


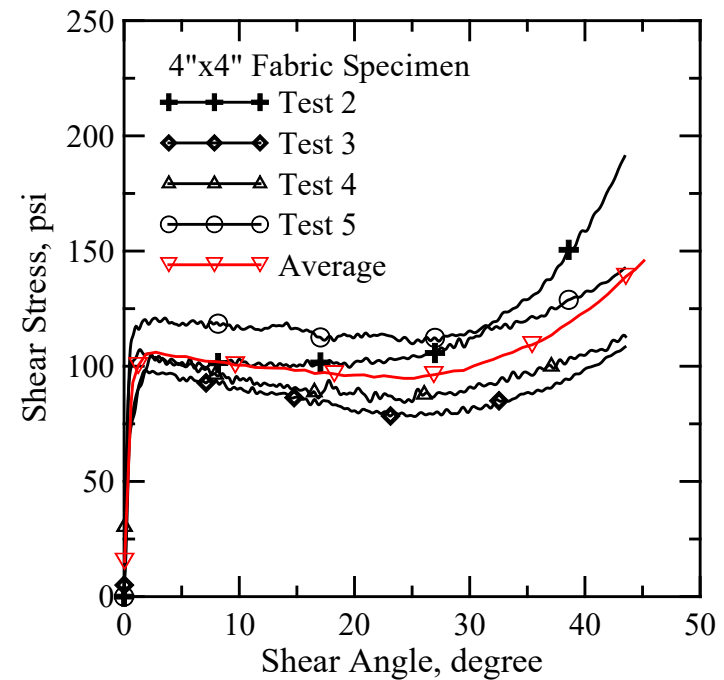
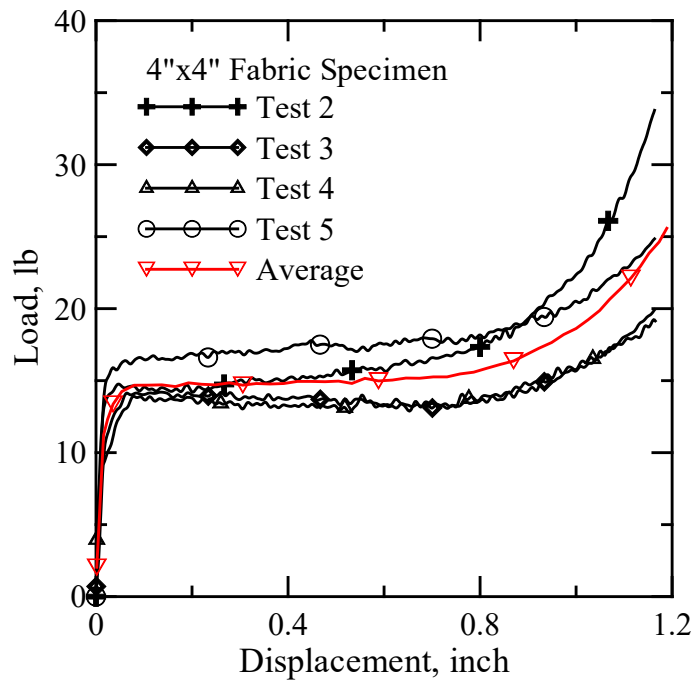
Three Size Tested:
3"x3", 4"x4", 5"x5"

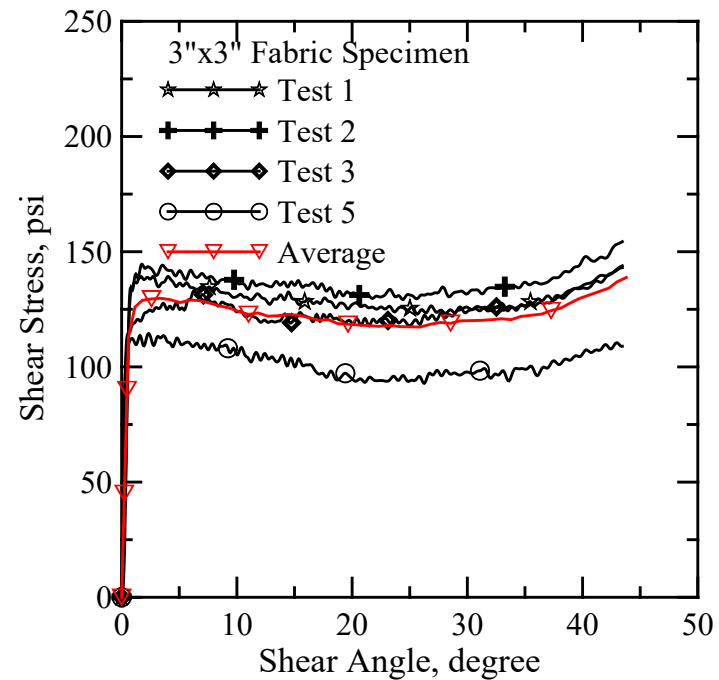
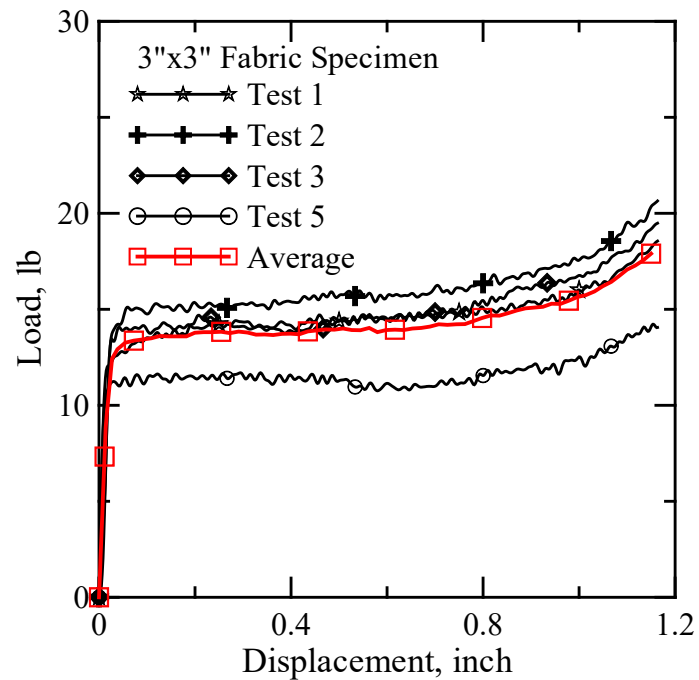
Video and Image Analysis (DIC)

34

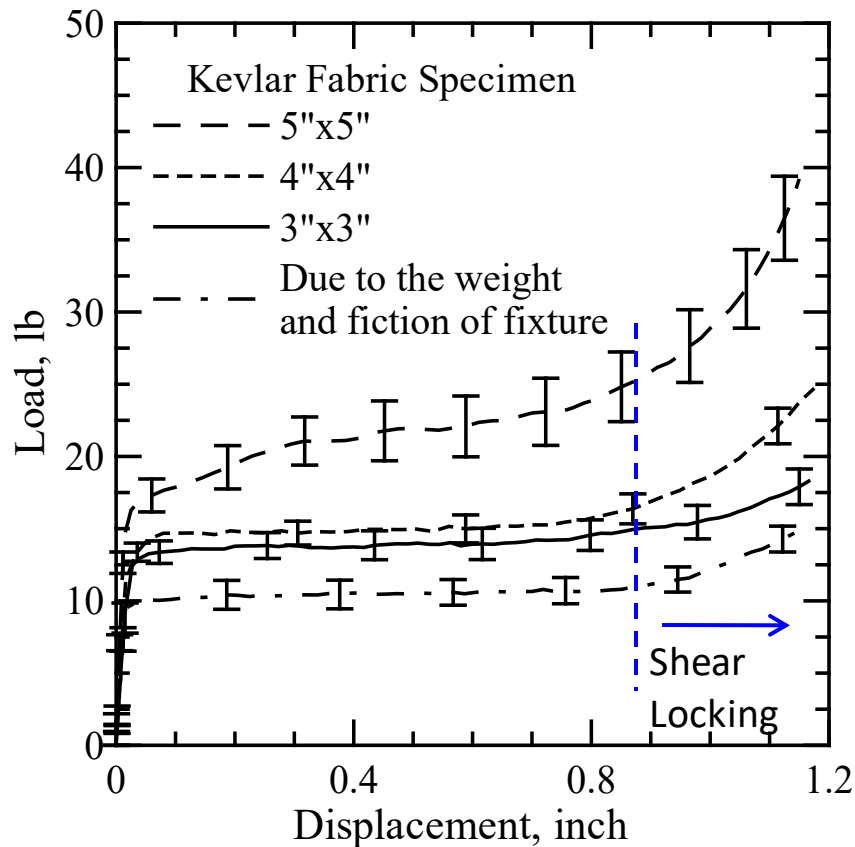




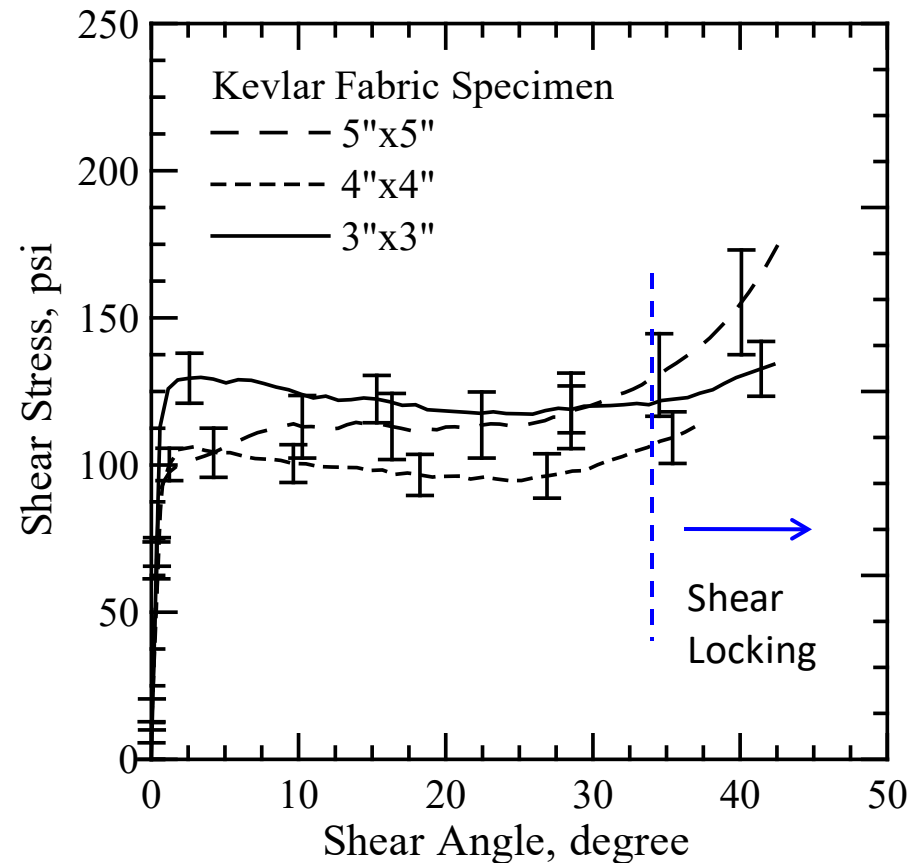




Picture Frame Testing Results



Load vs. Displacement

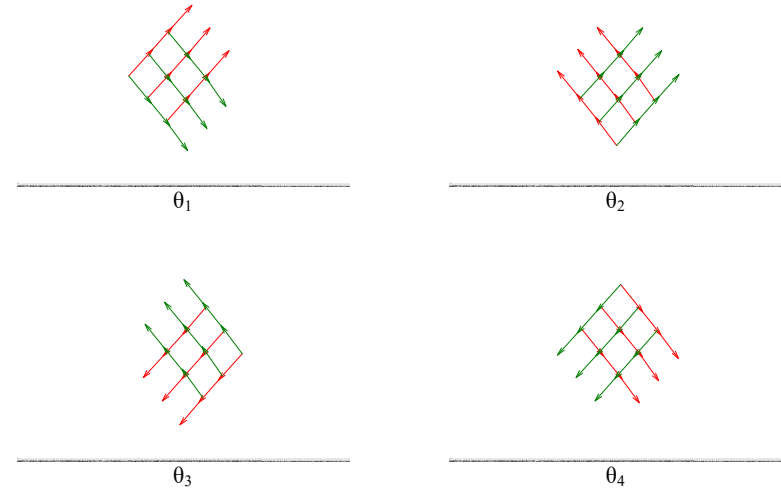
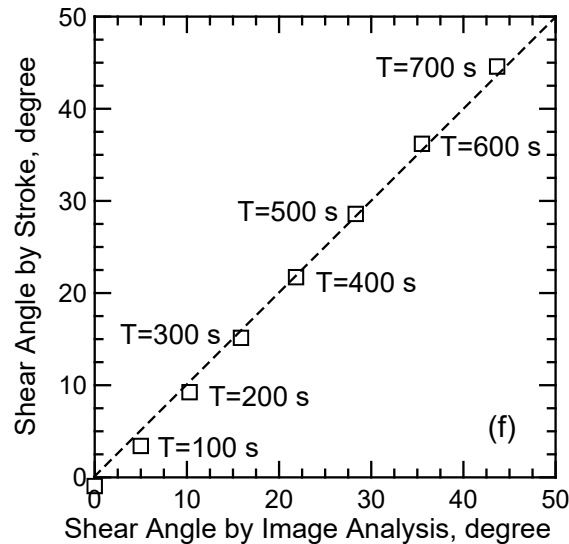


Shear Stress vs. Shear Angle

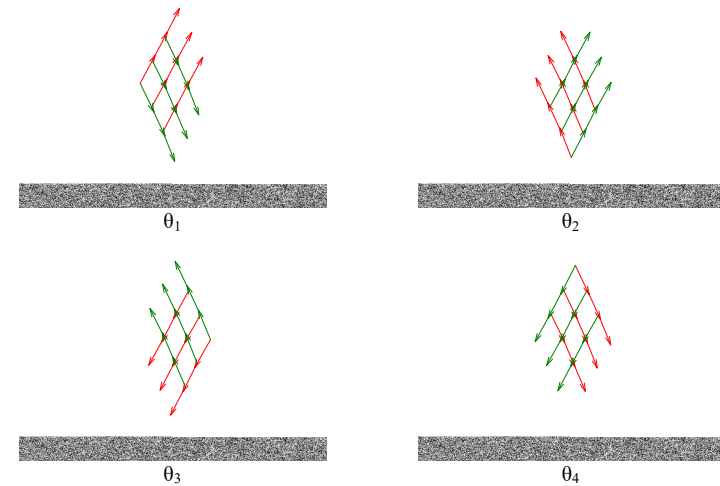


Chosen locations for the calculation of shear angle and X, Y displacement

Comparison of shear angles of image analysis with stroke measurement



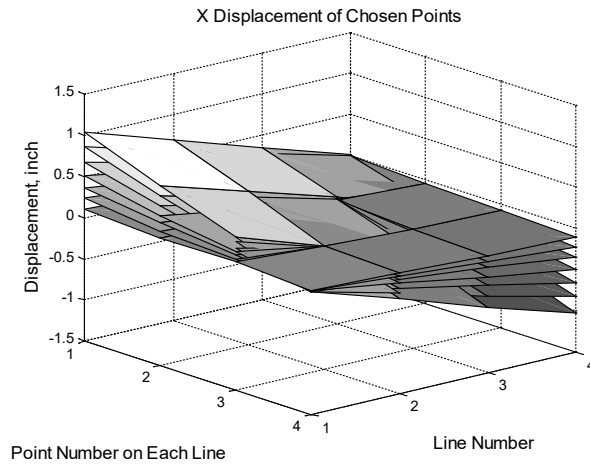
Shear angles at 100 sec



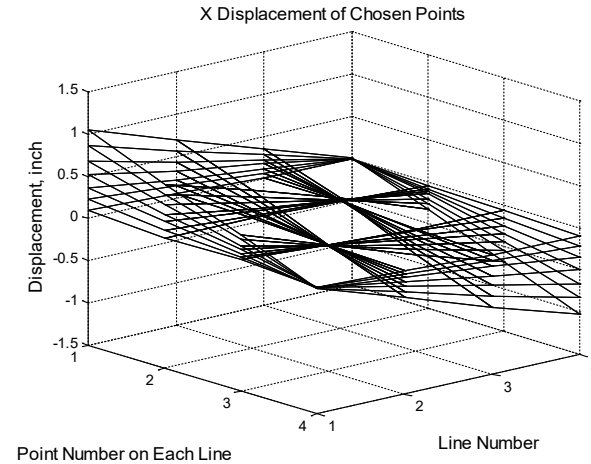
Shear angles at 300 sec

Image Analysis of Full-field Displacement

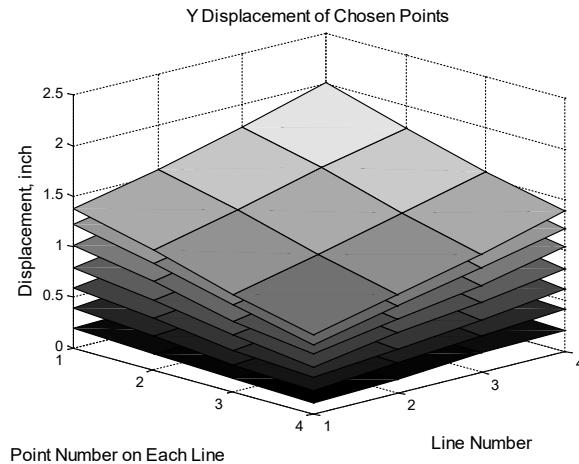
40



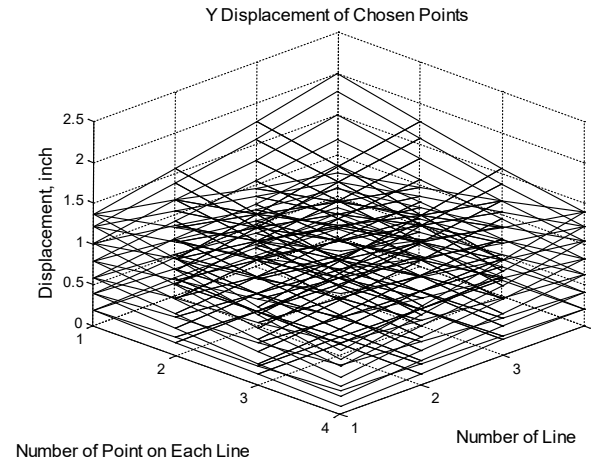
(a) X displacement



(b) X displacement

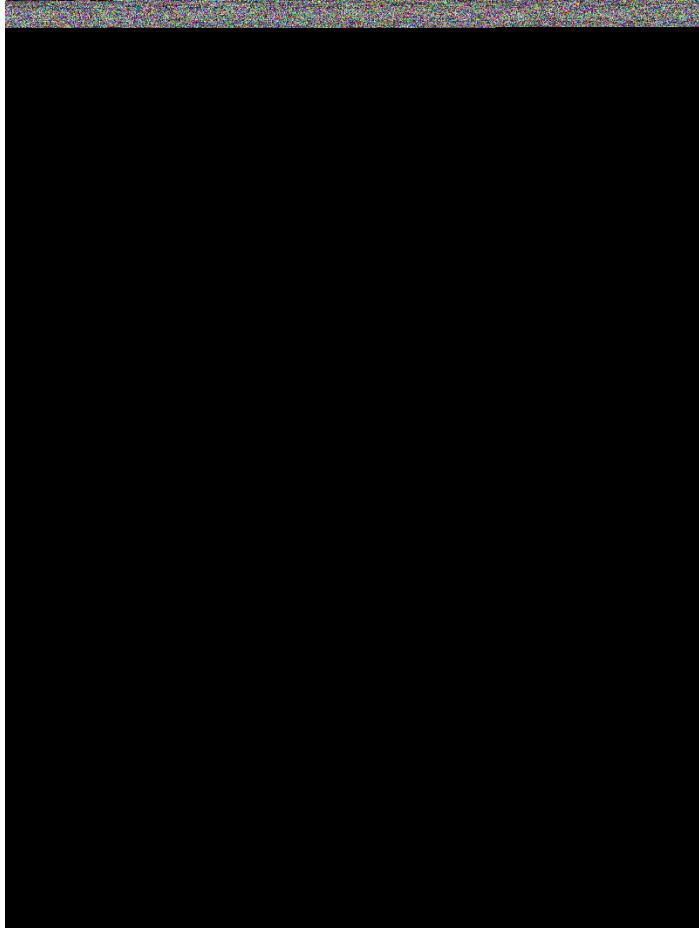


(c) Y displacement

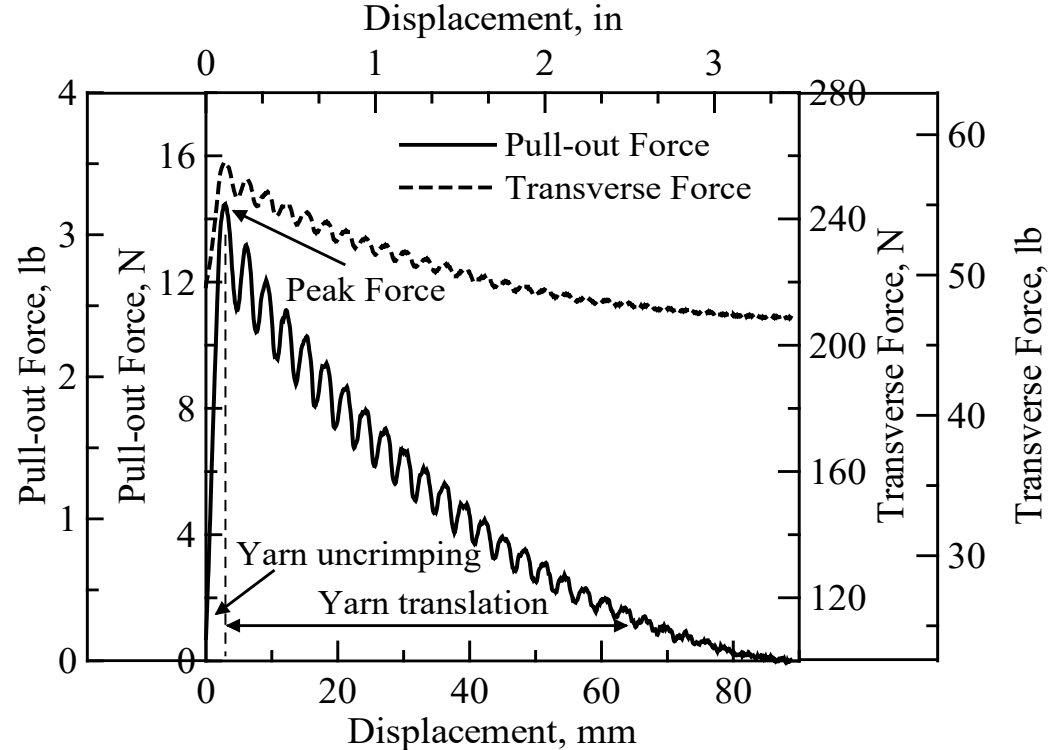


(d) Y displacement

Single Yarn Pullout



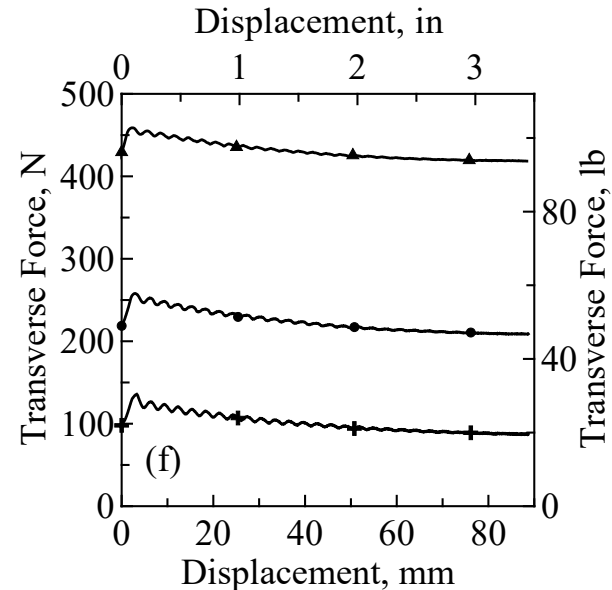
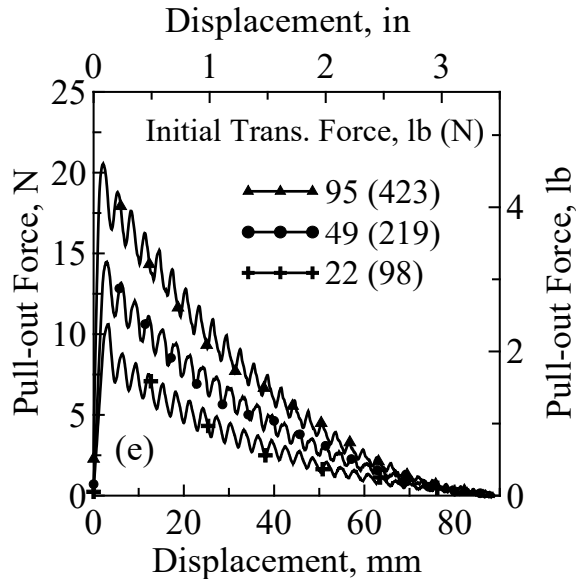
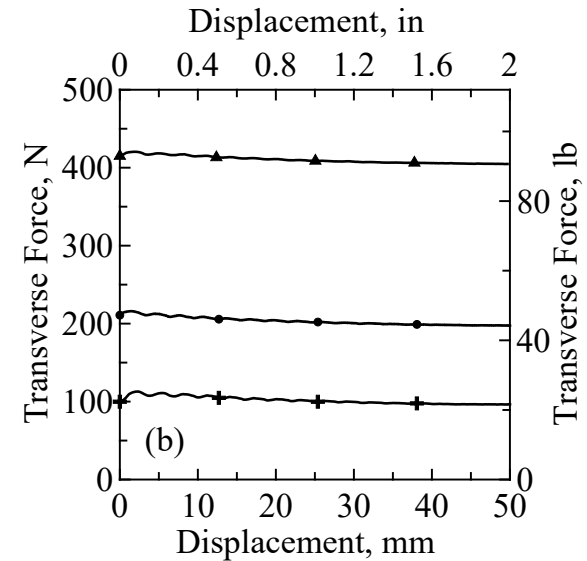
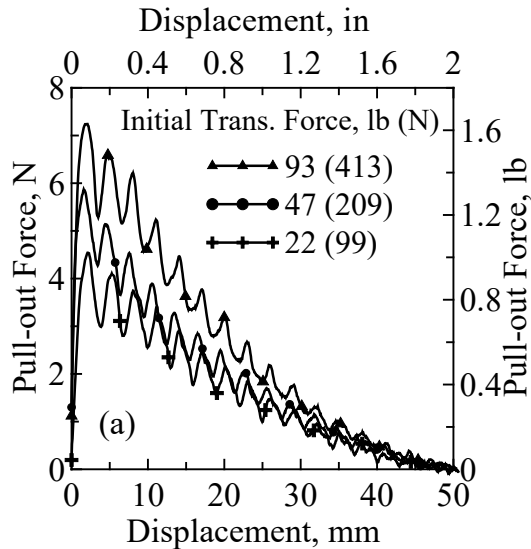
Typical Response during Yarn Pull-Out



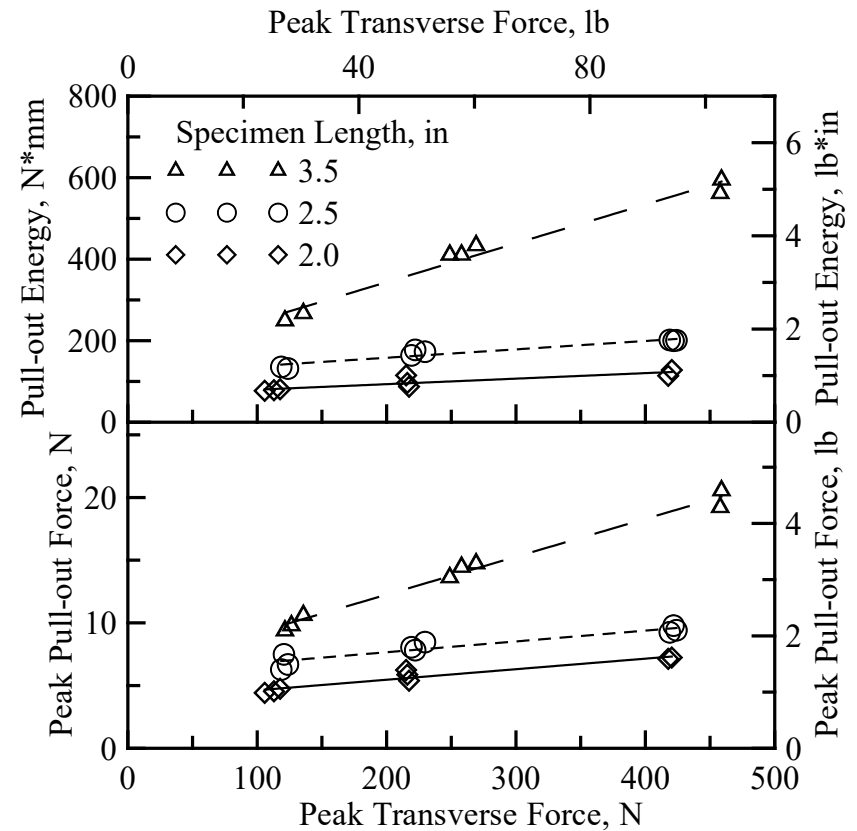
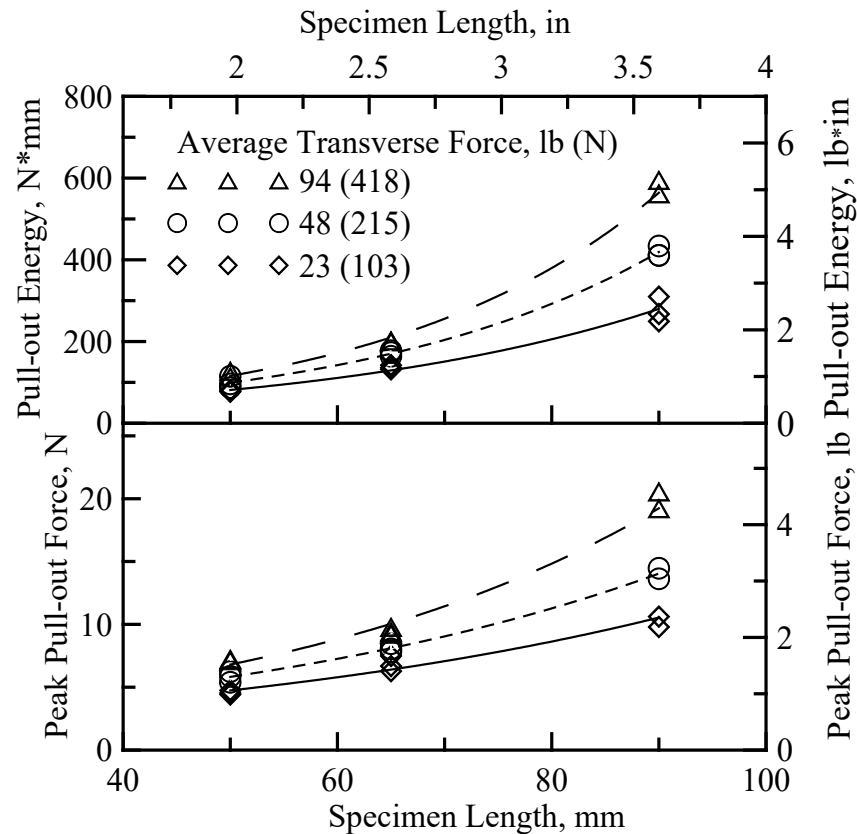
2" long x 3.5" wide

Representative Pull-out Curves

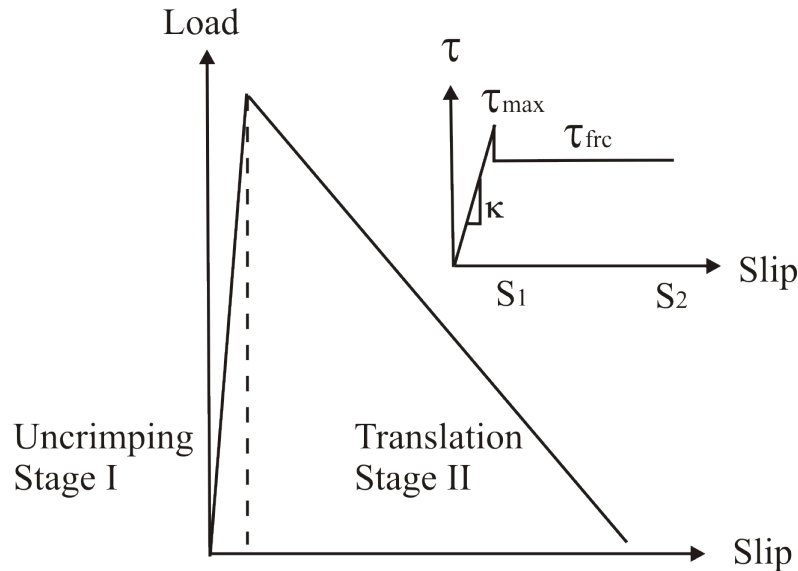
43



Effects of Specimen Length and Transverse Force



Analytical Model of Pull-out Behavior of Single Yarn

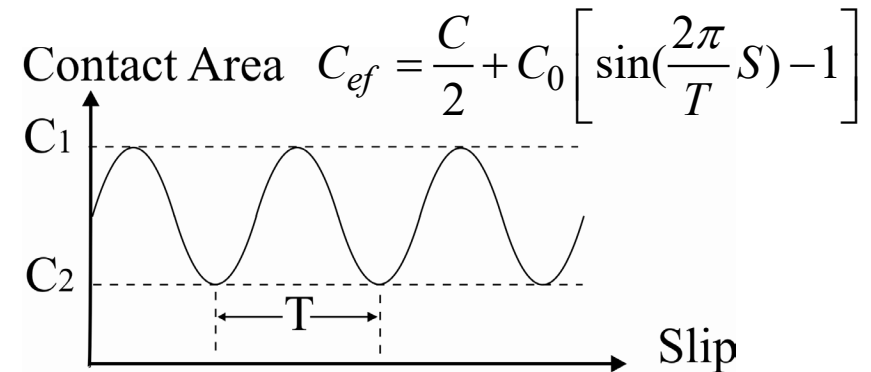


Pullout-Slip Response and Shear Strength Diagram

Shear Strength and Force Distribution along Yarn:
(a) Stage I (Uncrimping Stage), (b) Stage II (Translation Stage)

Reference: Sueki S., Soranakom C., Mobasher B., and Peled A. Pullout-Slip Response of Fabrics Embedded in a Cement Paste Matrix. Journal of Materials in Civil Engineering 2007; 19(9): 718-727.

Analytical Model of Pull-out Behavior of Single Yarn (continue)



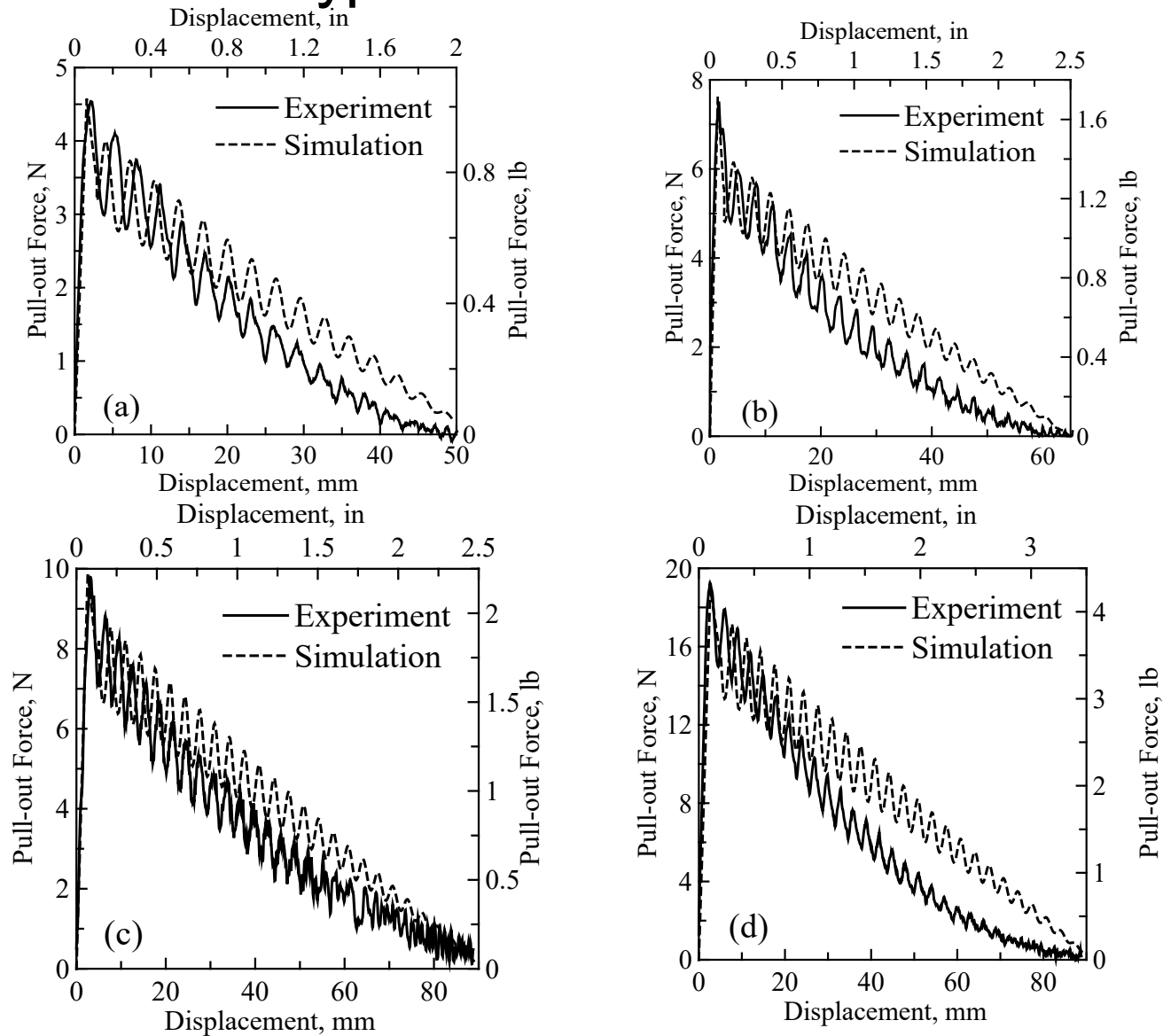
Change of Contact Area during Pull-out

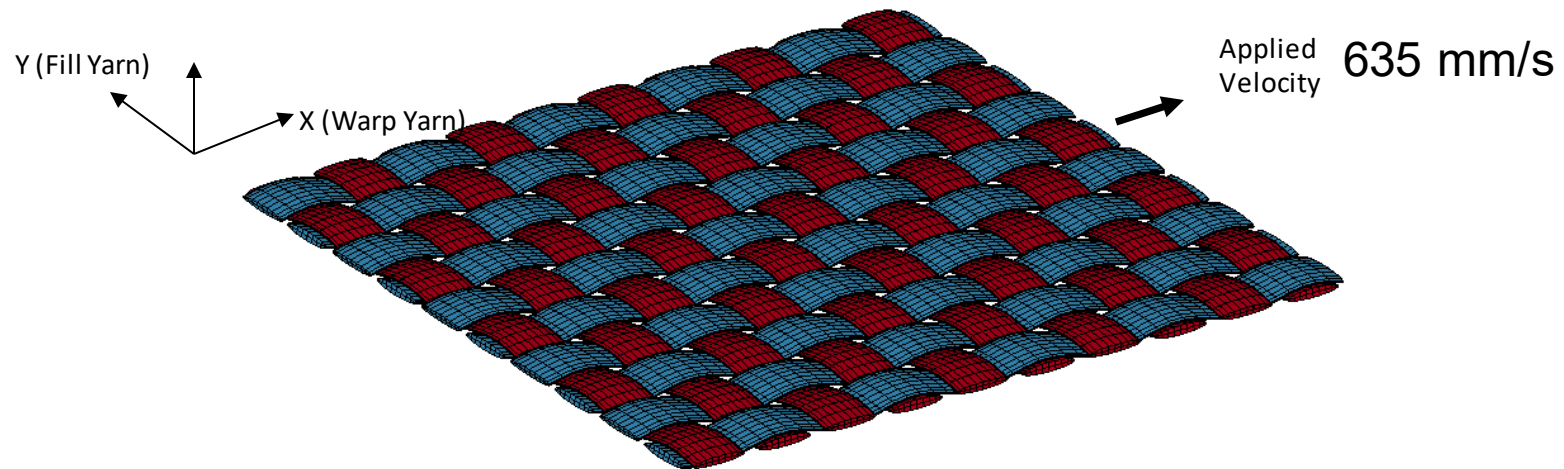


Direction	Elliptical Cross-section		Sinusoidal Path	
	Semi - major axis (mm)	Semi - minor axis (mm)	Period (mm)	Amplitude (mm)
Warp Yarn	0.655	0.085	2.997	0.122
Fill Yarn	0.585	0.089	2.982	0.096

Cross-sectional micro-image of (a) warp yarn and (b) fill yarn in the fabric

Typical Simulation Results





(1) Geometry: The 3-D model contains 10x10 yarns. (20,000 eight-noded solid elements and 36300 nodes).

(2) Materials: Orthotropic linear-elastic material data (GPa)

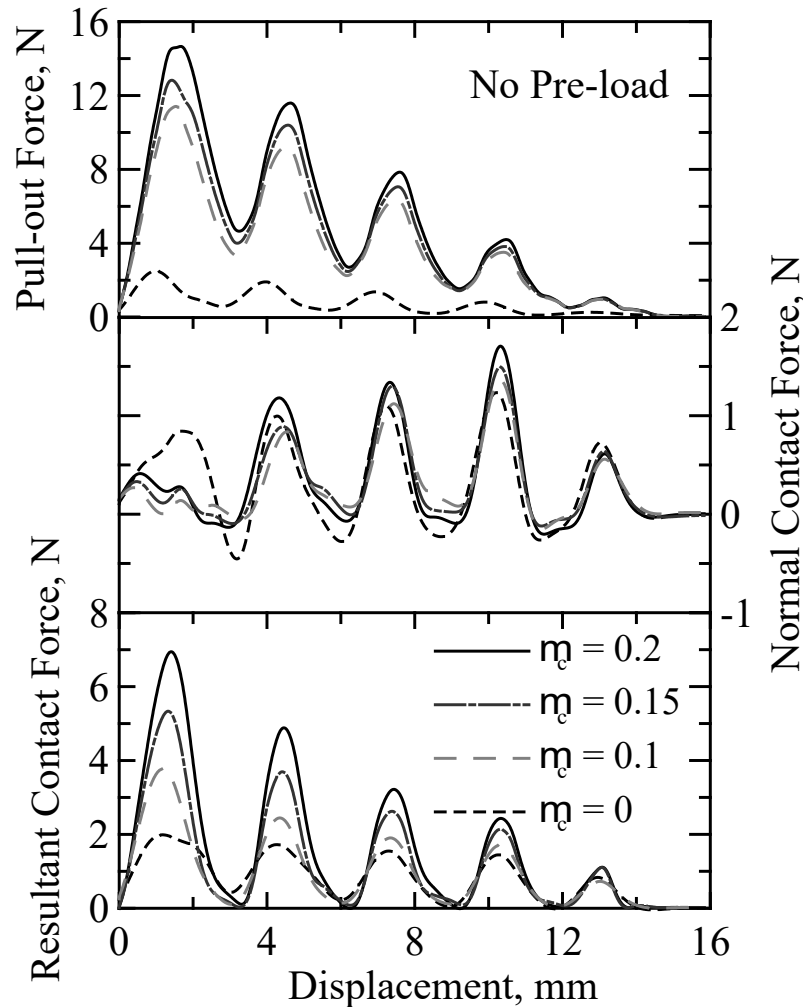
E_{11}	E_{22}	E_{33}	G_{12}	G_{13}	G_{23}	ν_{12}	ν_{13}	ν_{23}
72	3.6	3.6	3.6	3.6	3.6	0	0	0

(3) Boundary conditions:

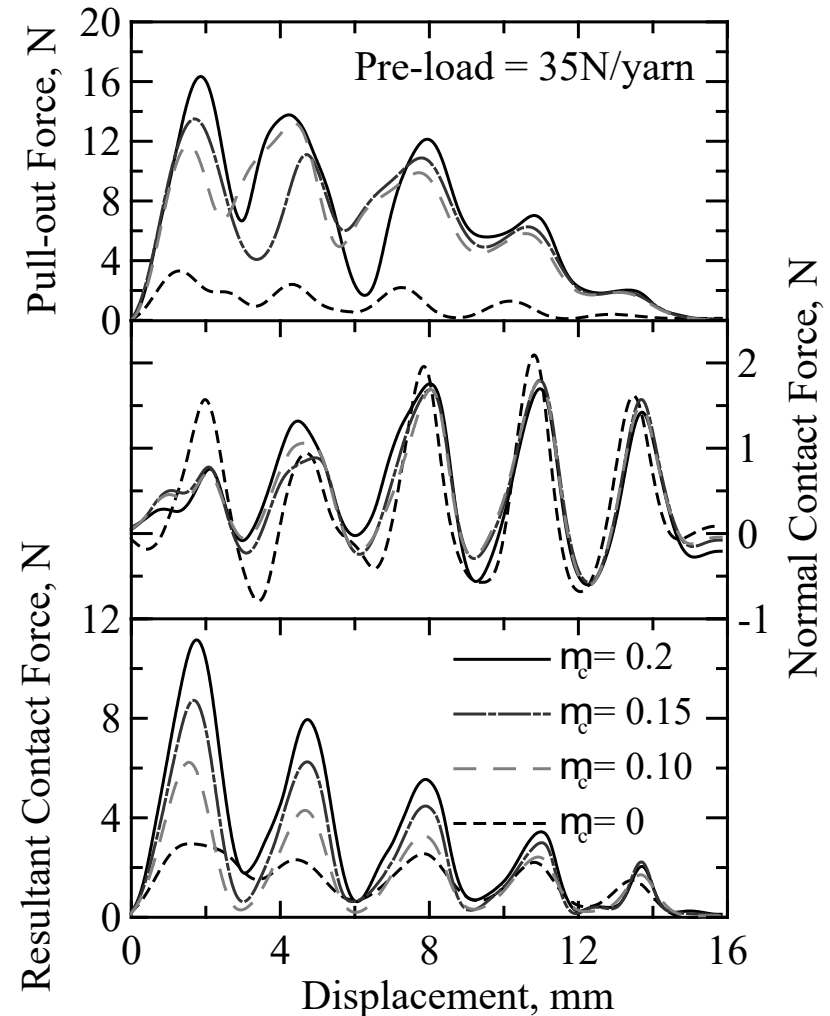
- two opposite edges clamped
- two opposite edges pre-loaded with certain amount of force

(4) Cases Study: case 1: $\mu_c = 0.2$; case 2: $\mu_c = 0.15$; case 3: $\mu_c = 0.10$;
case 4: $\mu_c = 0$

Simulation Results: pull-out force, normal force and resultant contact force vs. displacement

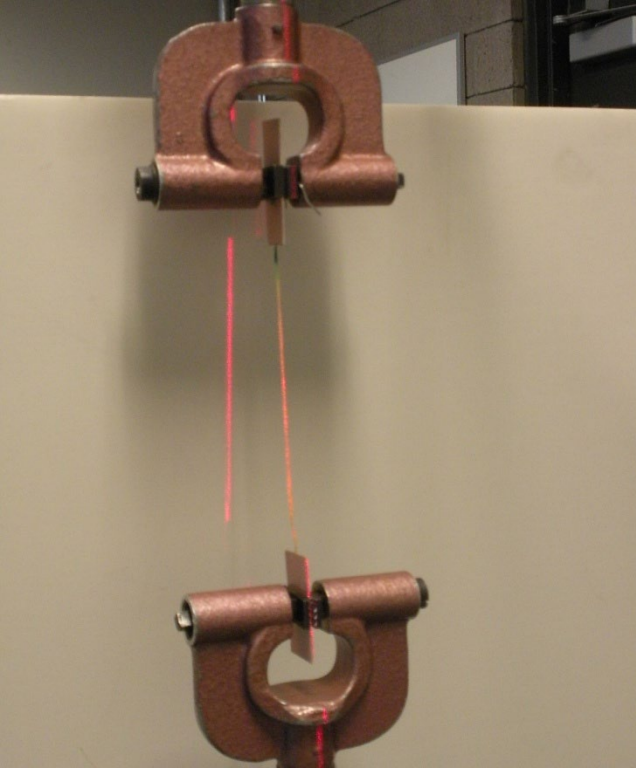


Response with Boundary Condition 1

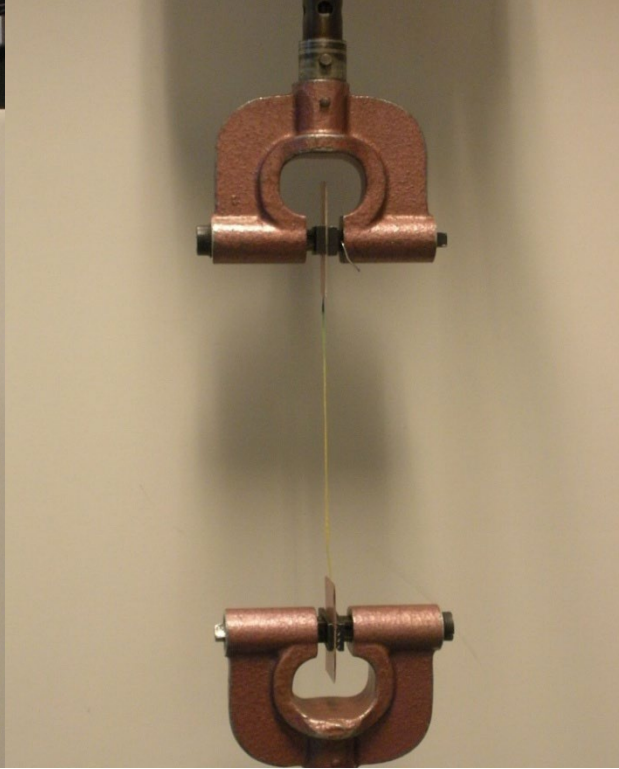


Response with Boundary Condition 2

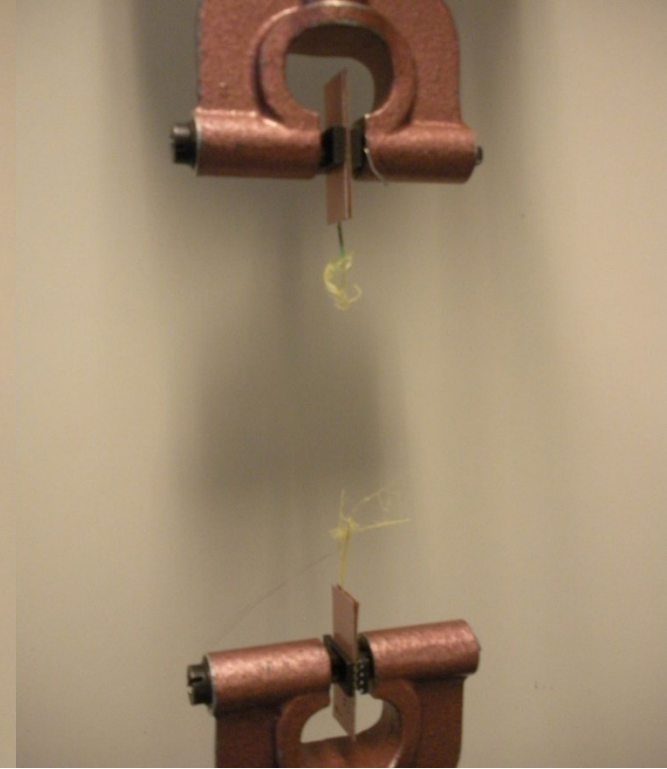
Single yarn quasi-static tension⁵⁰ test on virgin yarn



Specimen set-up and
alignment



Specimen during the test



Failed specimen

Specimen length and rate of loading ⁵¹

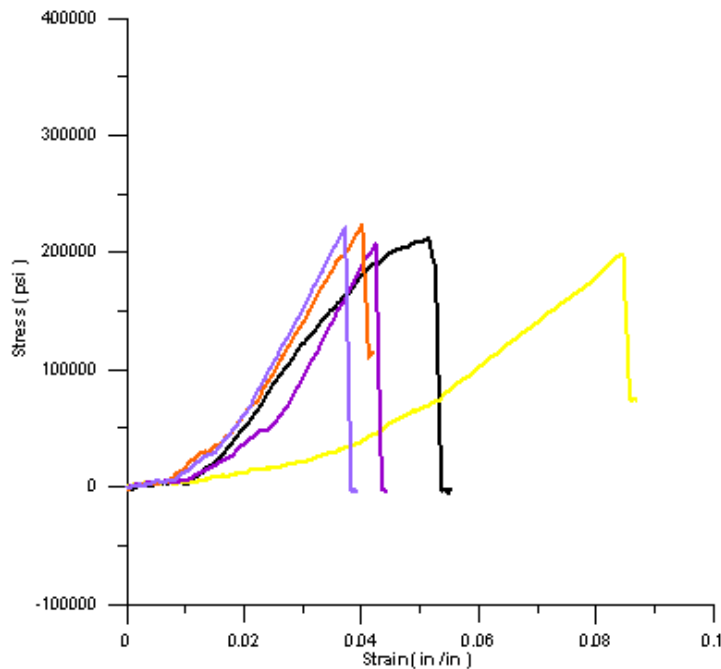
Specimen length (in)	Rate of loading (in / min)
2	0.05
5	0.125
8	0.2
11	0.275
14	0.35
17	0.425

Summary of Results

Specimen Length	Number of refined specimens	Maximum force (lb)	Maximum stress (psi)	Strain at maximum stress (in / in)	Modulus (psi)	Toughness (psi)
2	5	38.57	212629.39	0.0511	7420303.13	4063.06
5	8	49.61	272079.56	0.0306	11122443.57	3913.17
8	5	40.85	224592.10	0.0193	13346931.74	2181.62
11	7	46.53	254546.20	0.0187	15445026.17	2313.78
14	7	41.55	228536.30	0.0159	16076907.53	1849.78
17	5	37.70	204115.27	0.0135	16552122.81	1407.81

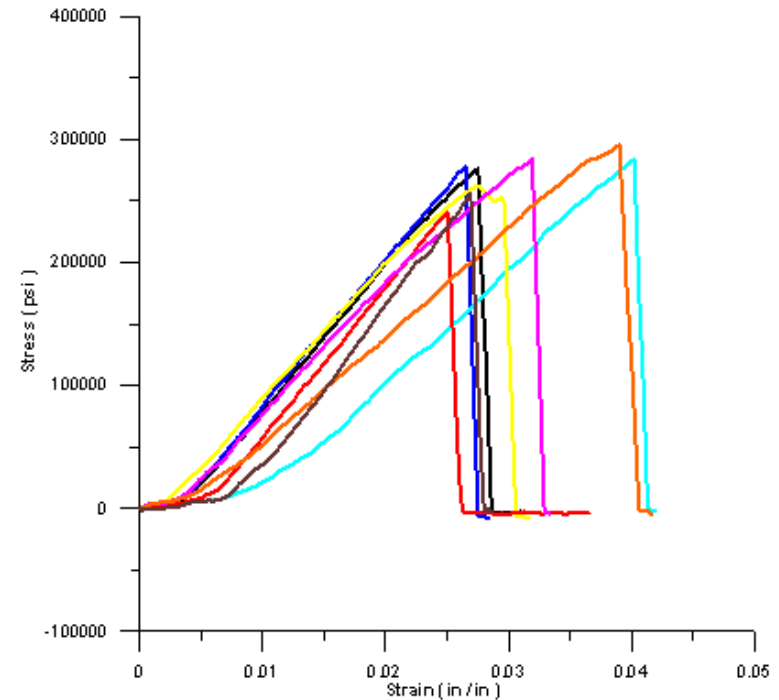
Response of 2,5 in specimen

Stress vs. Strain 2in specimen



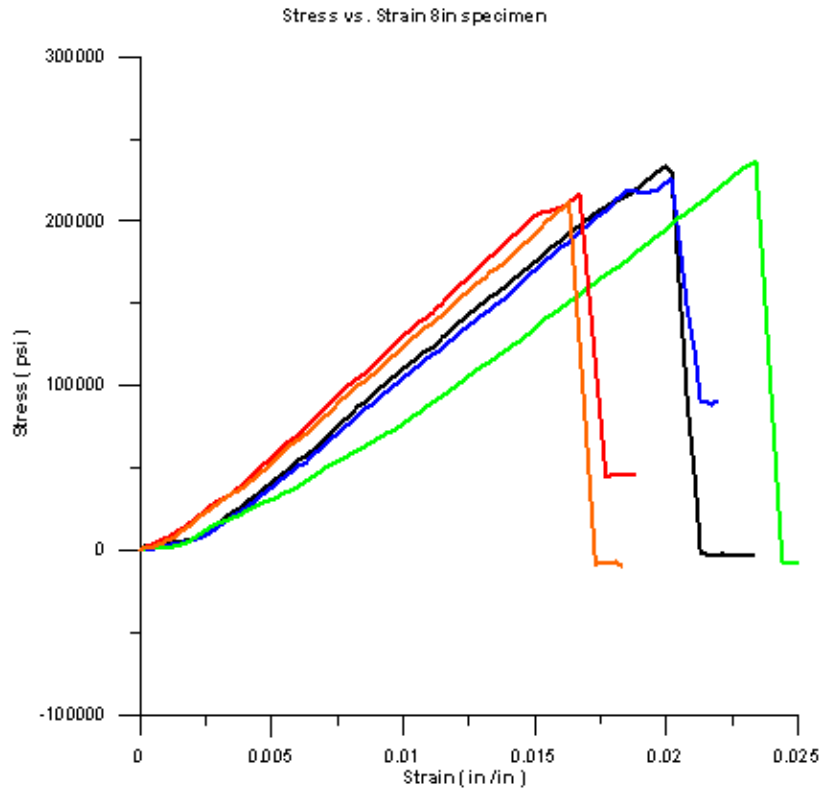
2 in specimen

Stress vs. Strain 5in specimen

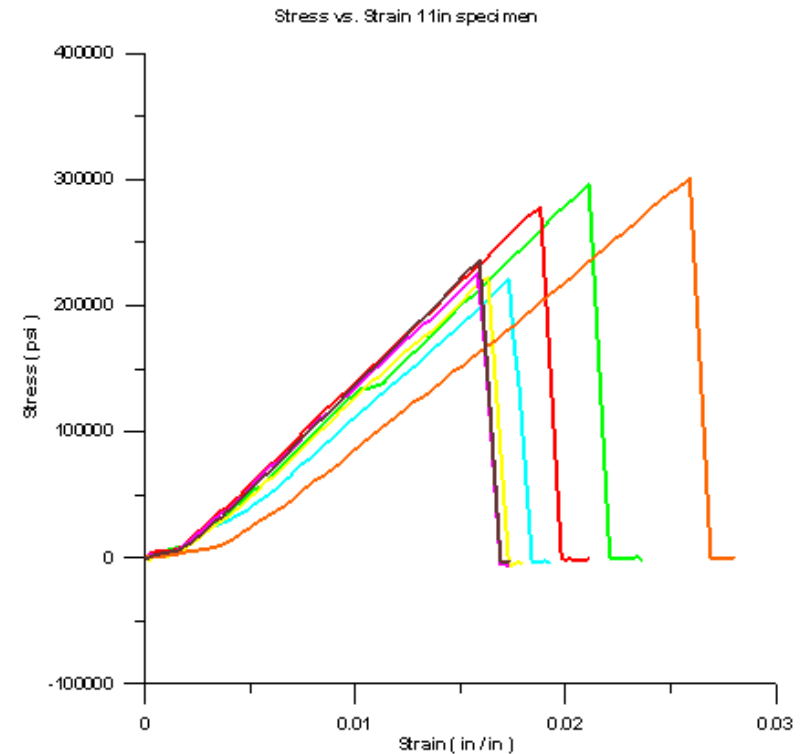


5 in specimen

Response of 8,11 in specimen

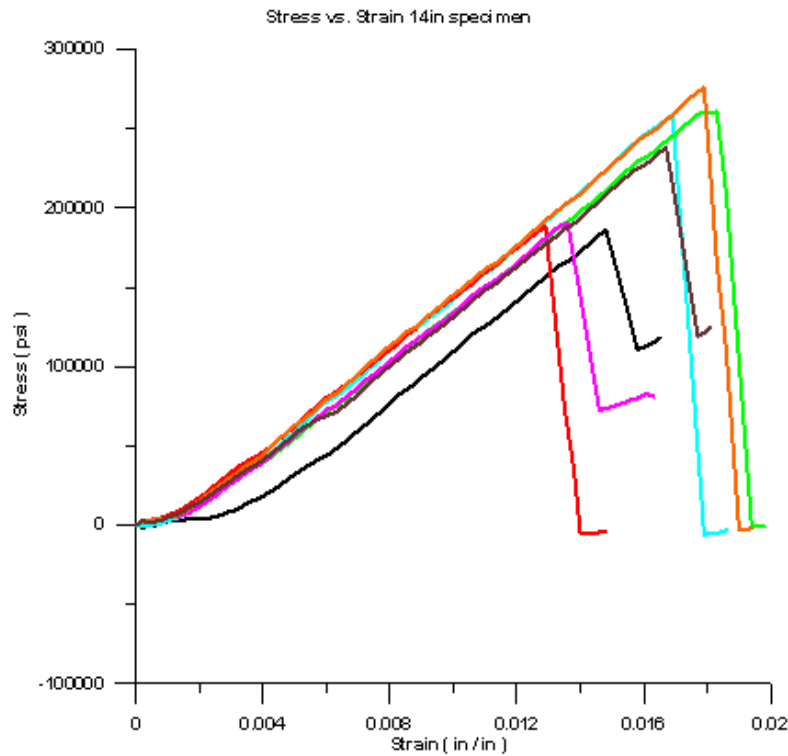


8 in specimen

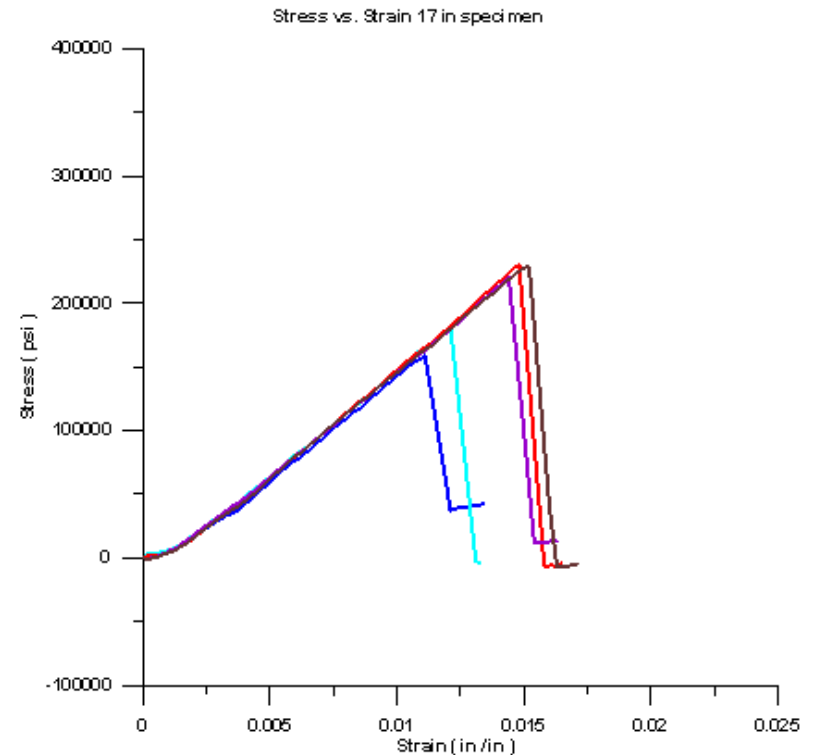


11 in specimen

Response of 14,17 in specimen

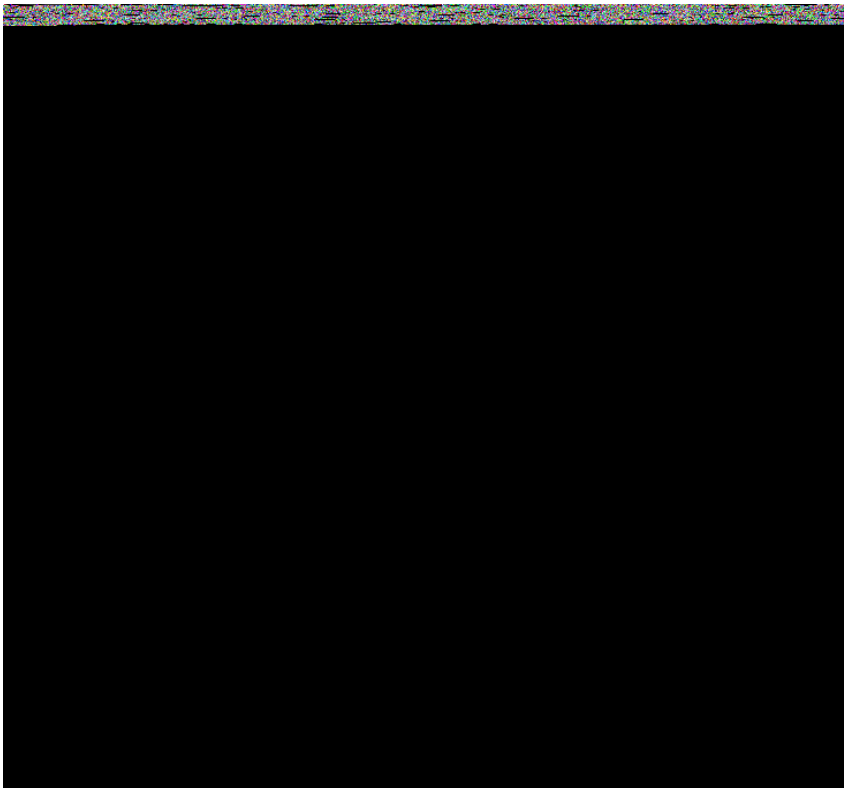


14 in specimen

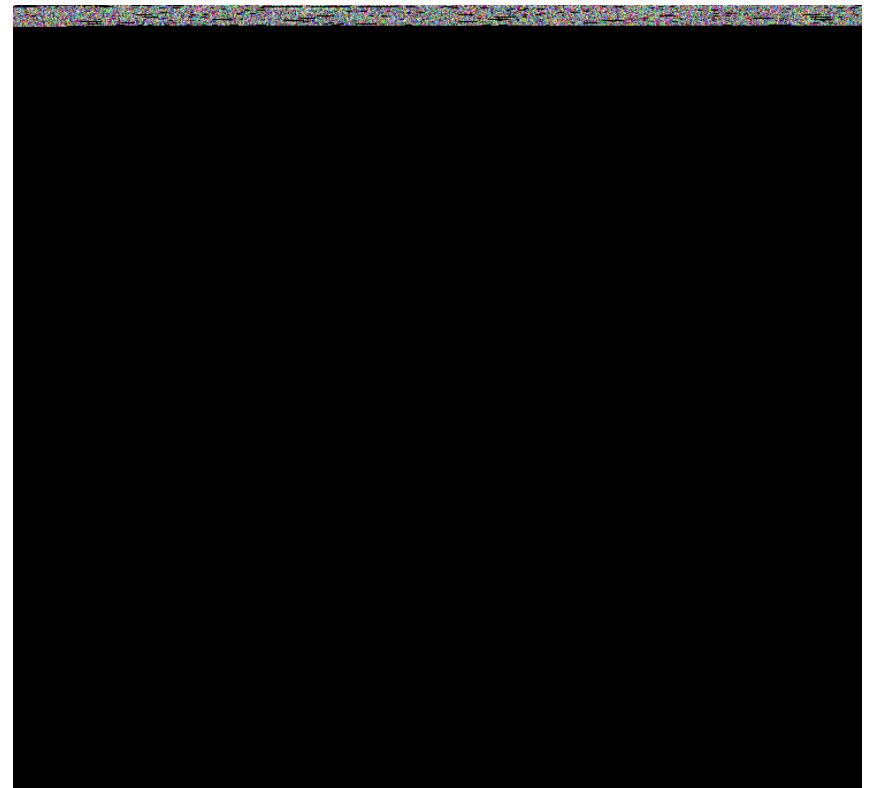


17 in specimen

Fill and Single yarn comparison



Warp and Single yarn comparison



Conclusions

- The static and dynamic material properties of Kevlar 49 Fabric and single yarn have been investigated. The results show that Kevlar 49 is sensitive to strain rate.
- Effective Poisson's ratio is nonlinear as a function of strain. Pre-load and sample size are factors in the determination of effective Poisson's ratio.
- The shear resistance in terms of shear force depends on the specimen size, i.e. larger specimen need more shear force to deform. The shear stresses of different size specimens are almost identical to each other before shear locking effect takes place.
- Pre-load, as well as friction plays a key role during to yarn pull-out from fabric.

Future Plan

- Design and manufacture a shear frame which can apply pretension and measure the change of tension during shear deformation to study the influence of pretension on the shear resistance.
- Conduct more tests with different loading rates to study effect of shear rate/velocity on the shear resistance.